Principles of Data Visualization

Dr. Jared Hutchins
AAEA Data Visualization Challenge
Webinar

Why have a webinar on principles?

For about 6 years, I have been teaching a data science class at the University of Illinois.

While I always taught the coding skills, I later realized that the principles of design are almost as important.

AI has made the coding part easier, allowing us to spend ***more time on the design part!***

Most of this presentation is based on this paper

Applied Economics Teaching Resources

Teaching and Educational Methods

Data Visualization in Applied Economics Instruction and Outreach

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JEL Codes: A2, C8, Q1, Y1
Keywords: data science, data visualization, Python, R programming

Abstract
This article highlights the critical role of data visualization in applied economics education and outreach. We first outline some general principles for teaching graph literacy and data visualization principles in and out of the classroom. We then discuss the mechanics of visualizing data—collection, preparation, and visualization—with an emphasis on how instructors can teach each step using the R and/or Python statistical environments. We ultimately contend that the requisite skills for successful data visualization are indispensable for students trained in today's agricultural and applied economics programs to communicate their research effectively.

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   a. Requires you to think carefully about what sort of story you are telling.
   b. What do you want to tell your audience in ***one sentence?***

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   a. Requires you to think carefully about what sort of story you are telling.
   b. What do you want to tell your audience in ***one sentence?***

2. Succinctly summarize data.
   a. Tables sometimes do not intuitively communicate results.
   b. Our brains are geared to sometimes take in information visually.

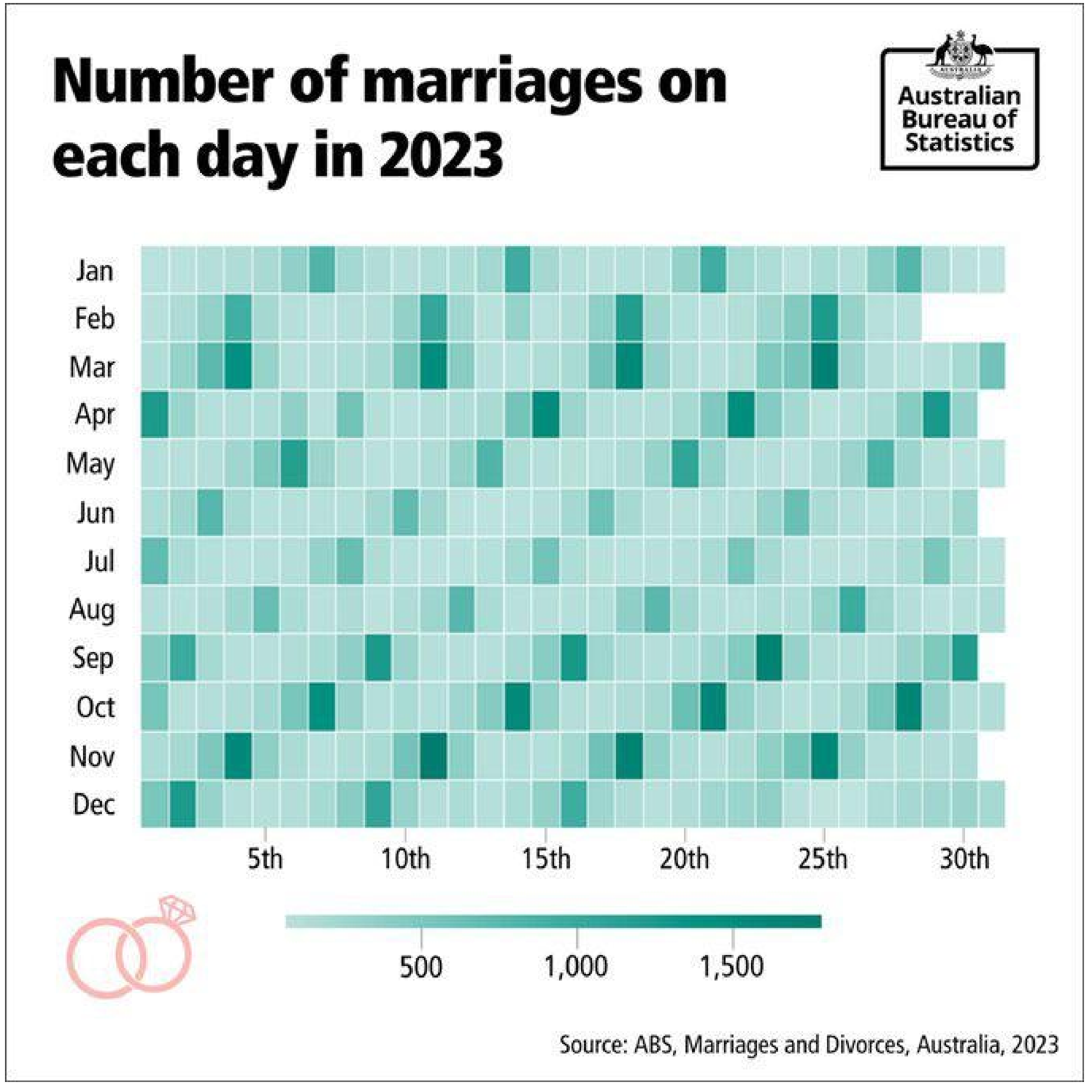
Know what the message is!

Data visualizations are ***not objective: they are always made with the intention of communicating something.***

If we know what we are communicating, the point of design is to help communicate this message.

The worst and most ineffective viz are made when we don’t know what we are trying to communicate.

Describe this graph in one sentence

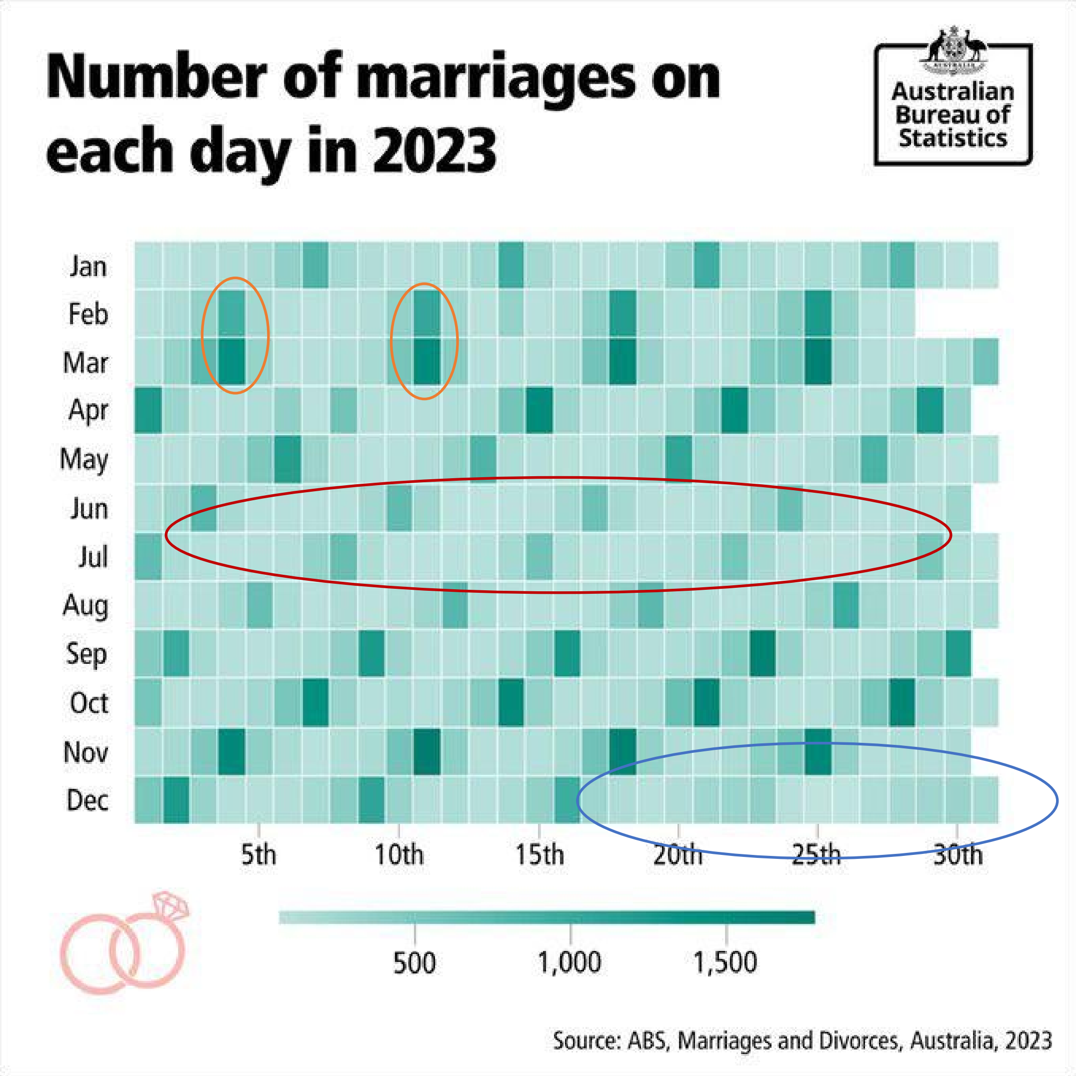


Describe this graph in one sentence

People get married on weekends

People get married less during the (Southern Hemisphere) winter

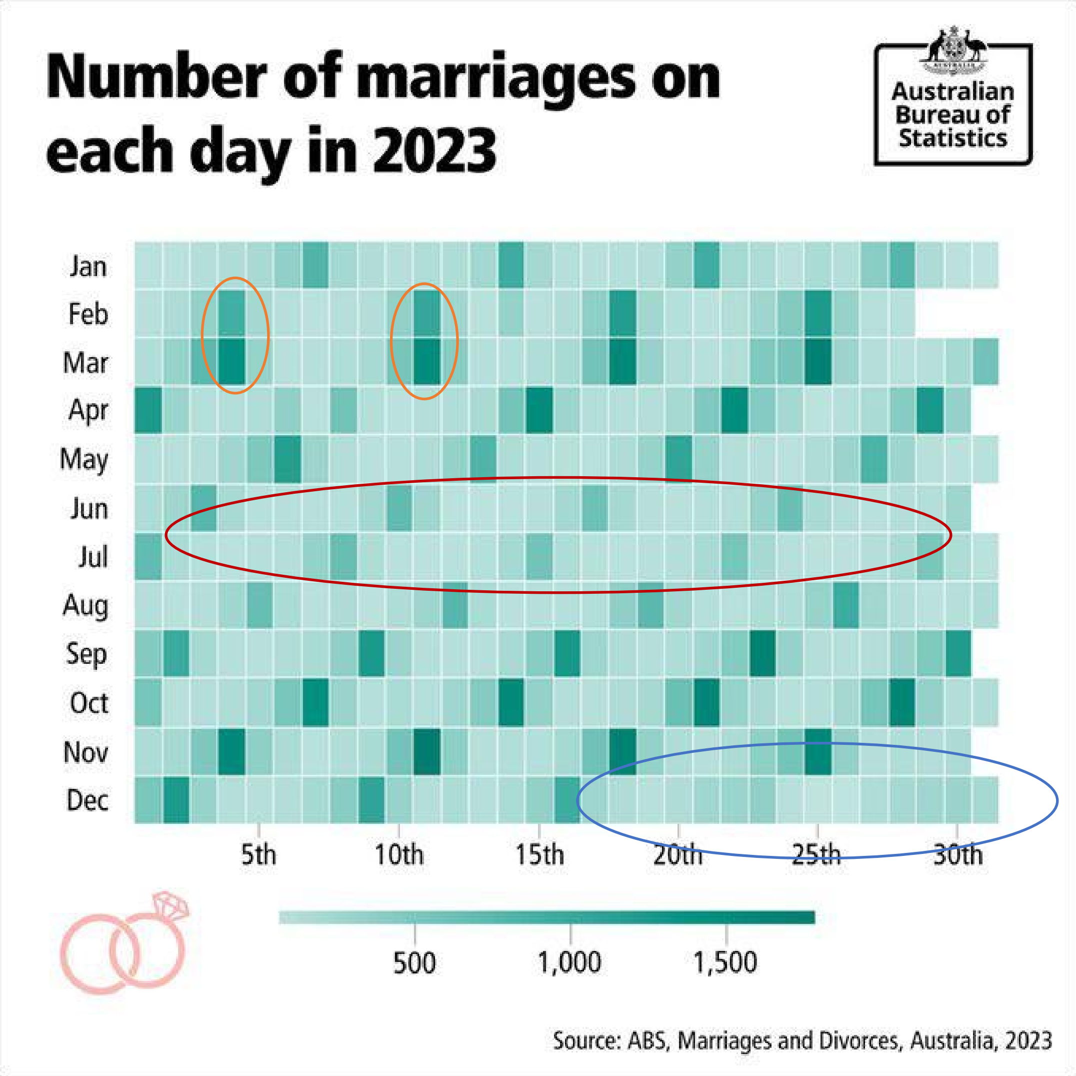
People don't get married during Christmas



Describe this graph in one sentence

In one sentence:

People tend to get married on the weekends, but not during winter or Christmas.



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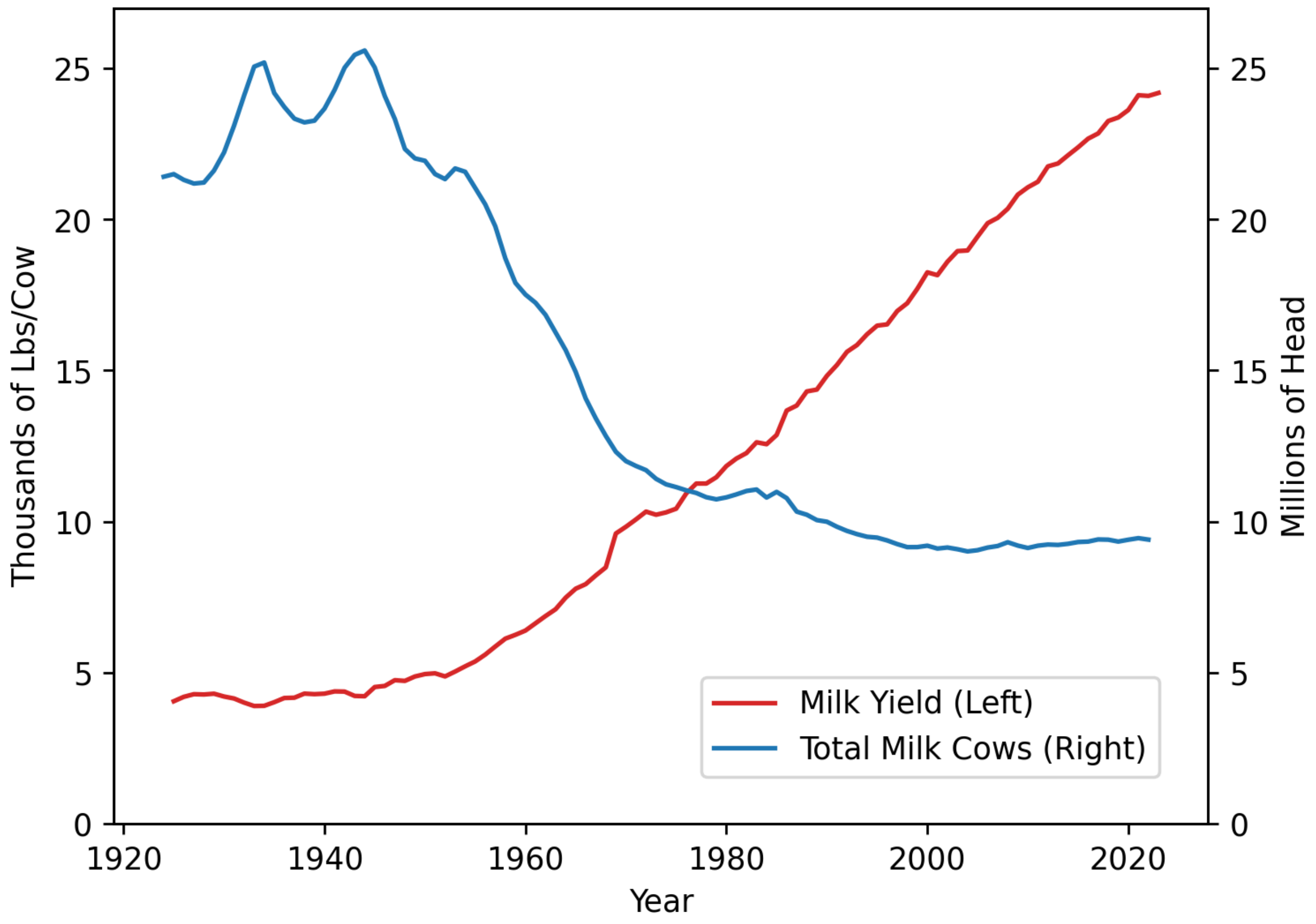
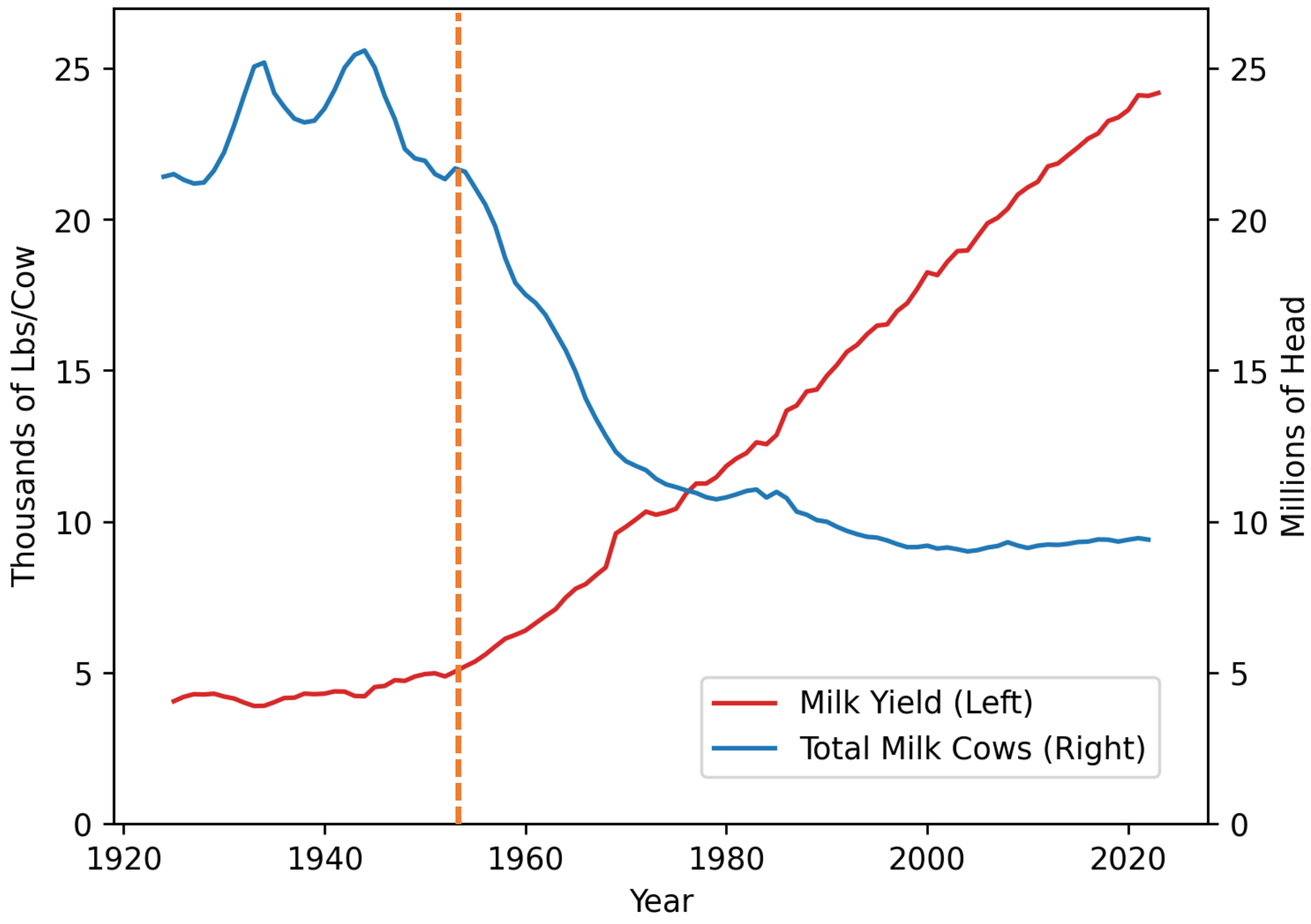


Describe this graph in one sentence

Milk yield has grown almost linearly.

Number of cows plummeted from about 22 million cows to a steady state of about 9 million.

Both these trends started around 1950.

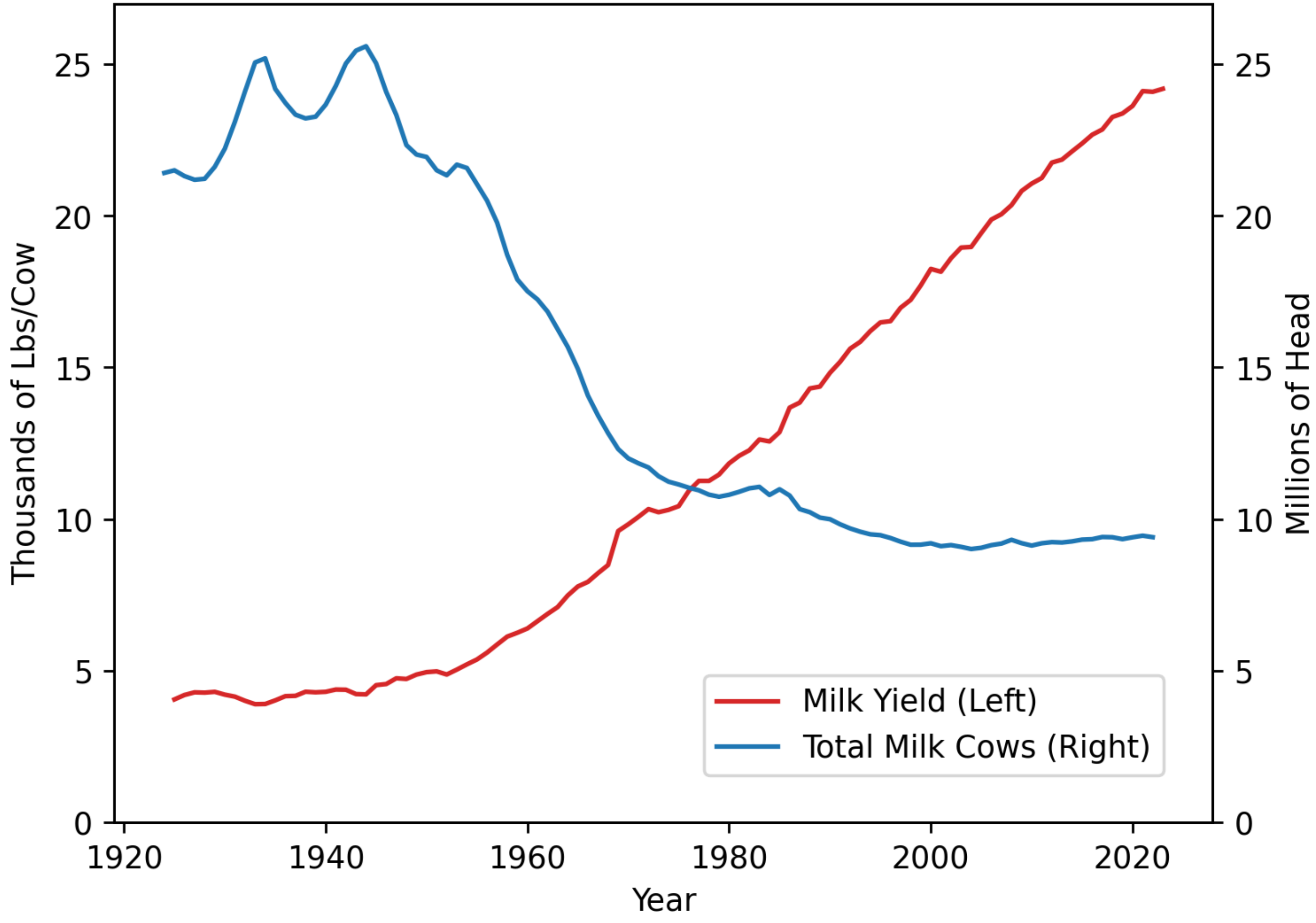


Describe this graph in one sentence

In one sentence:

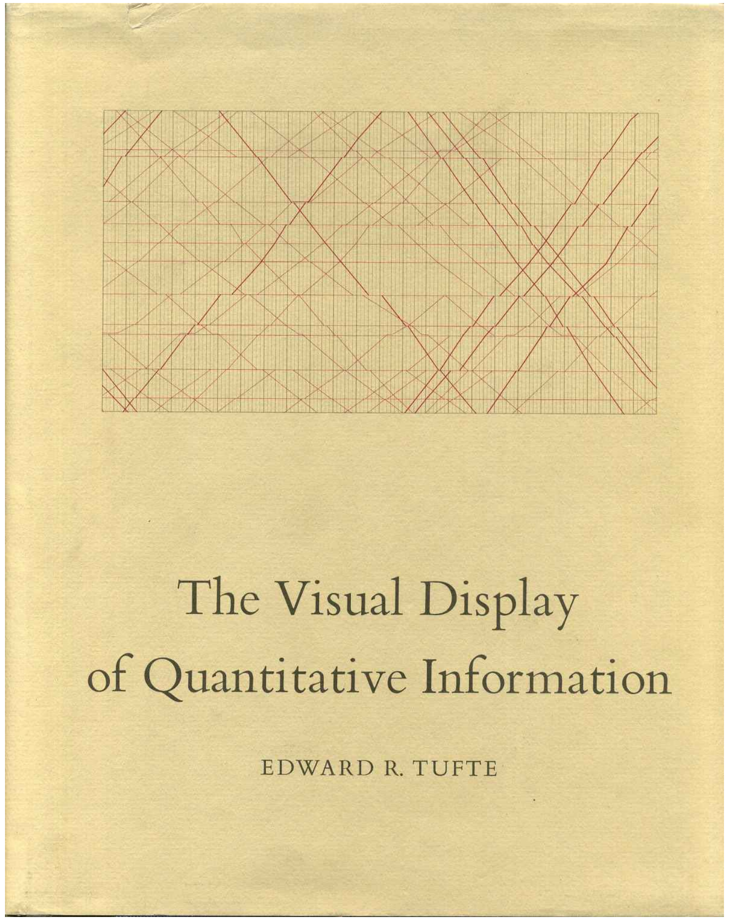
Starting in 1950, the dairy industry has drastically transformed into a more productive sector that uses less cows.

Even simpler: ***more milk, less cows!***

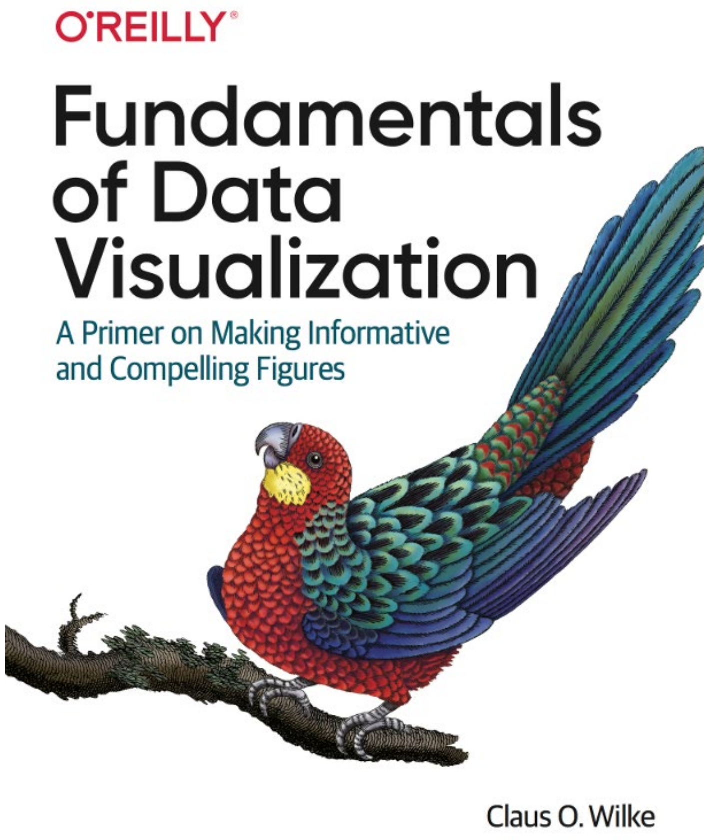


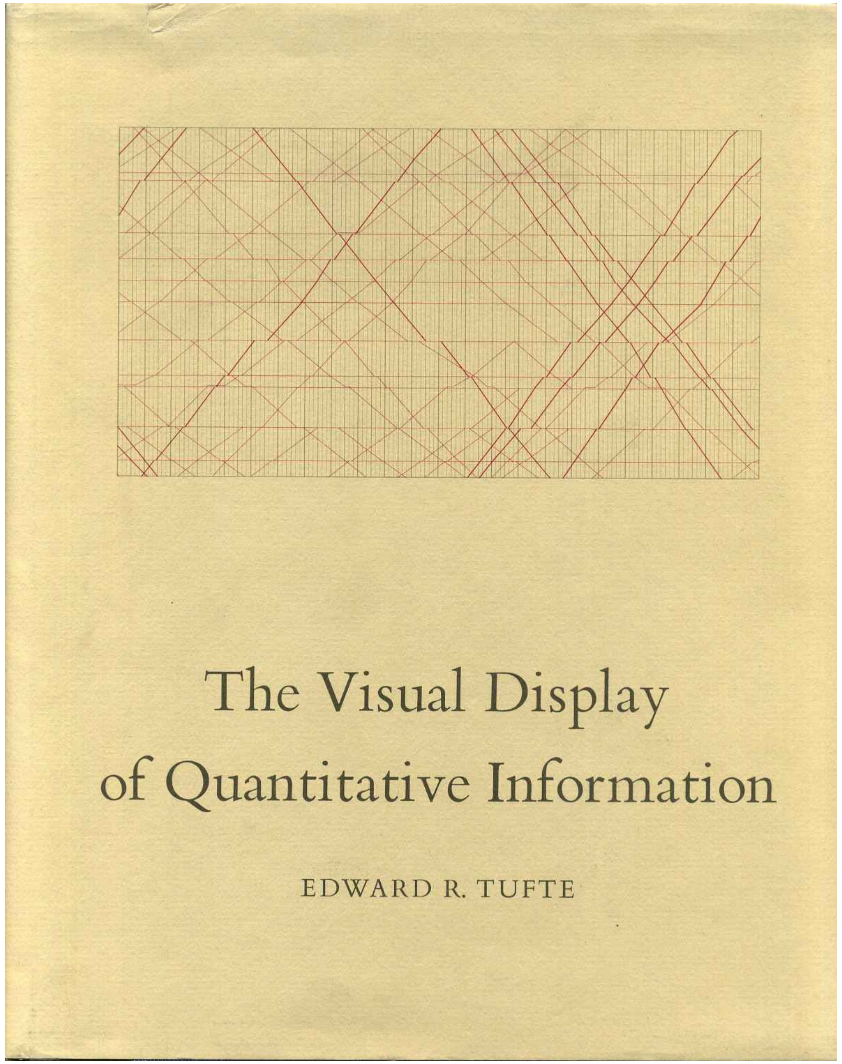
This webinar will focus on two areas

1) Lessons on Design from Tufte

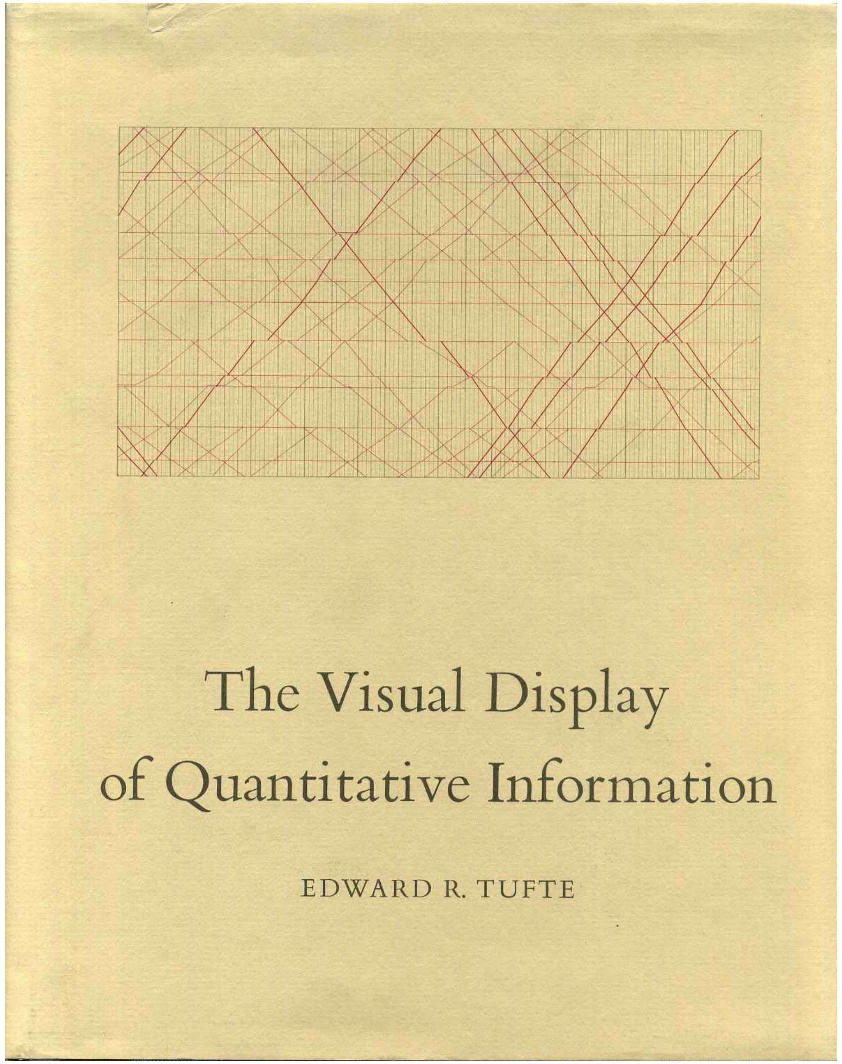


2) Lessons on Color from Wilke





Data Visualization Principles: Lessons from Tufte



Edward Tufte's book is considered a seminal text of data visualization principles.

While he has some strong (and controversial) opinions, his ***5 principles*** are useful to consider:

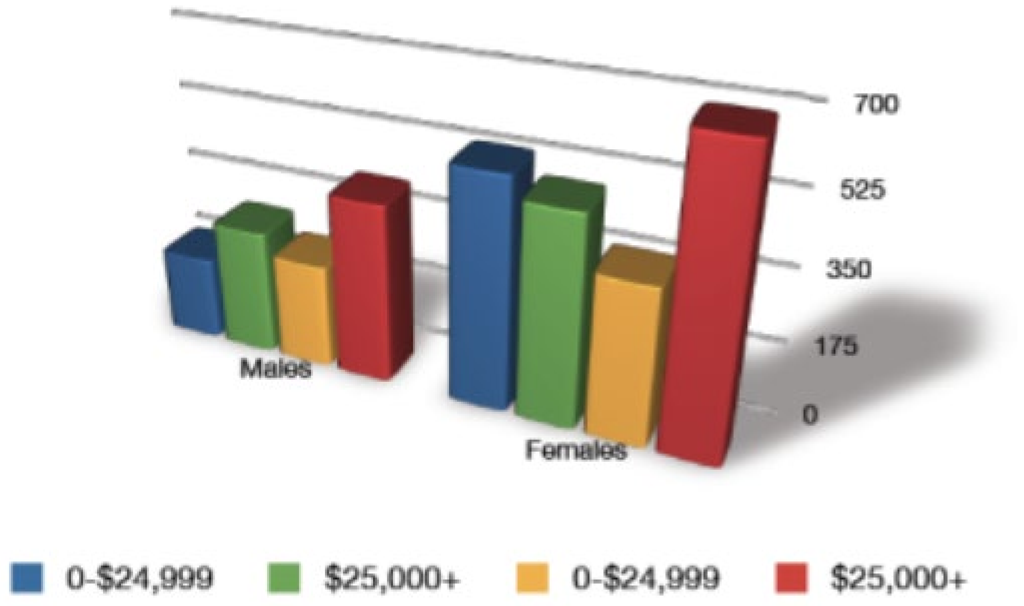
1. Above all else, show the data
2. Maximize the data-ink ratio
3. Erase non-data ink
4. Erase redundant data-ink
5. Revise and edit

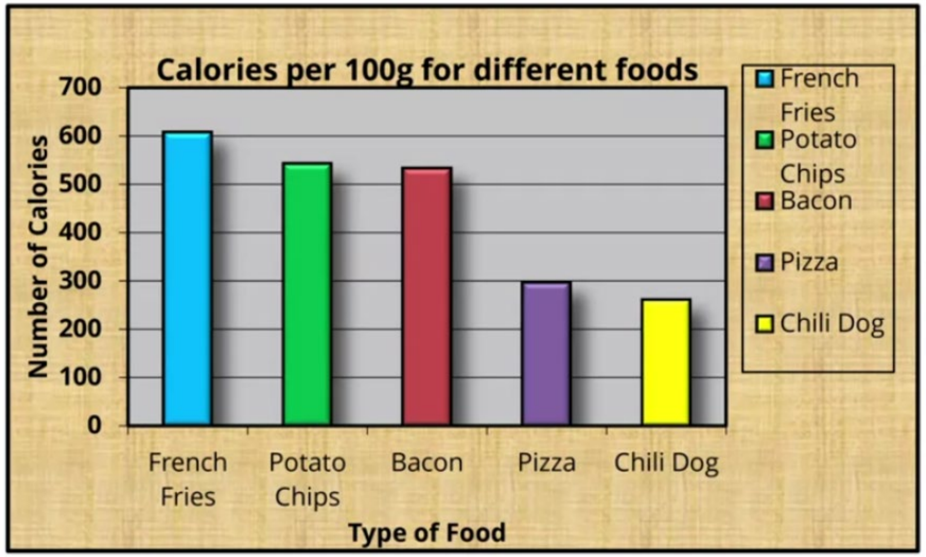
Data-ink ratio

What part of your figure is actually showing data?

Tufte specifies “data-ink” as the ink in your figure actually showing data.

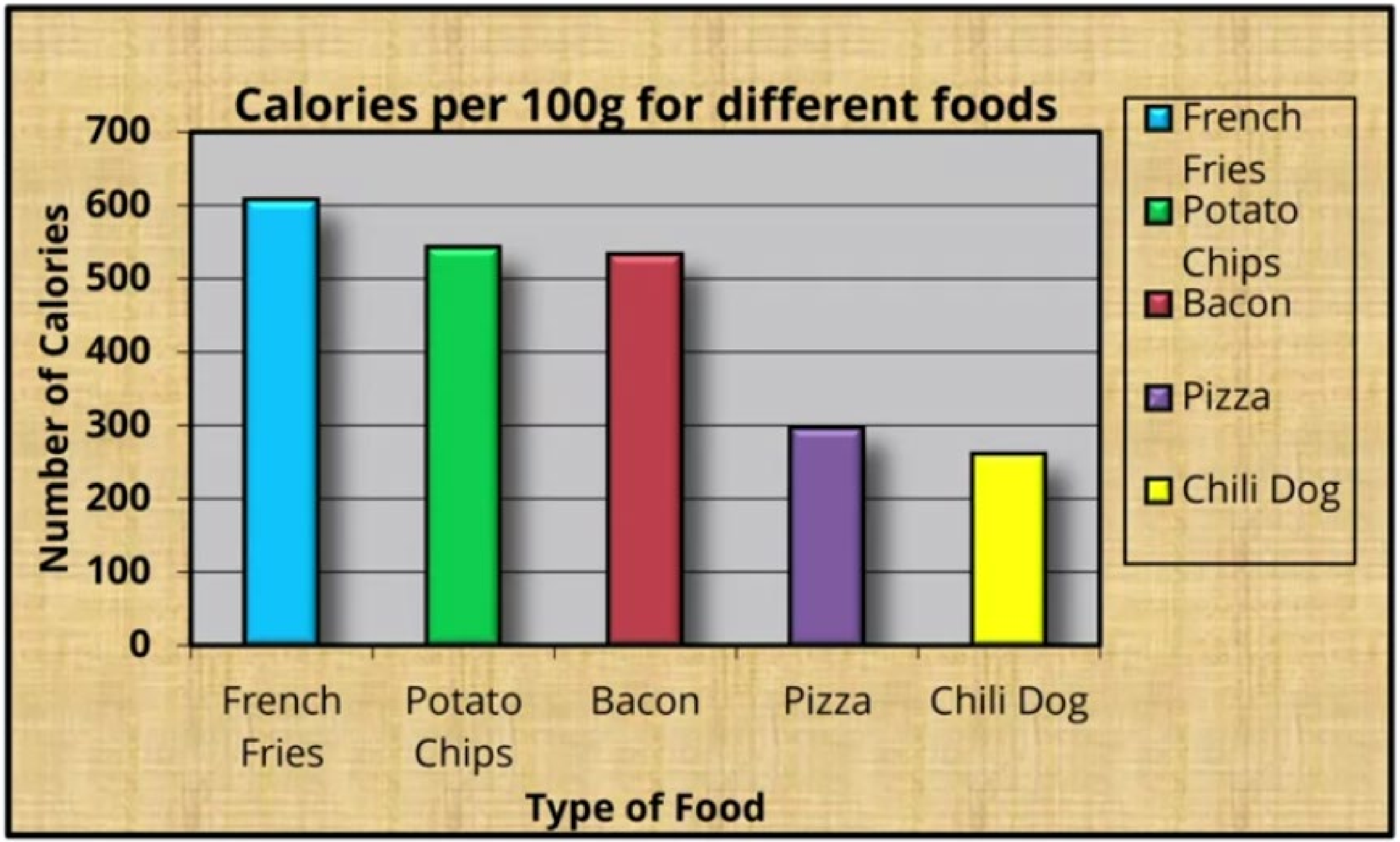
Data ink ratio = data-ink/total ink





Data-ink ratio

Which of the parts of this graph below are actually necessary?

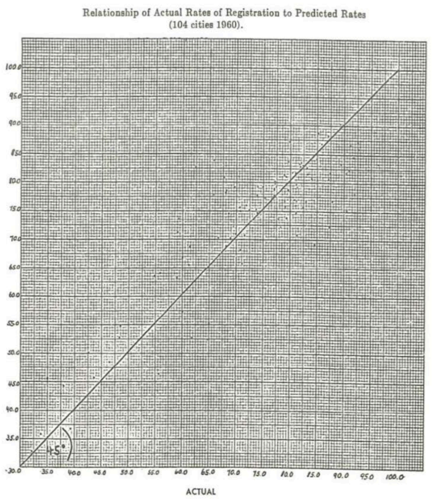


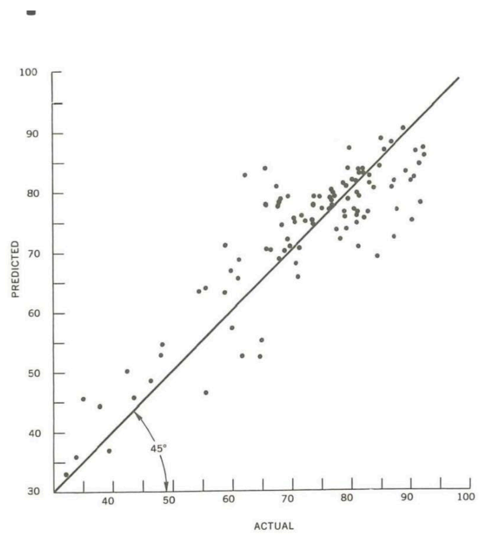
Erasing “chart junk”

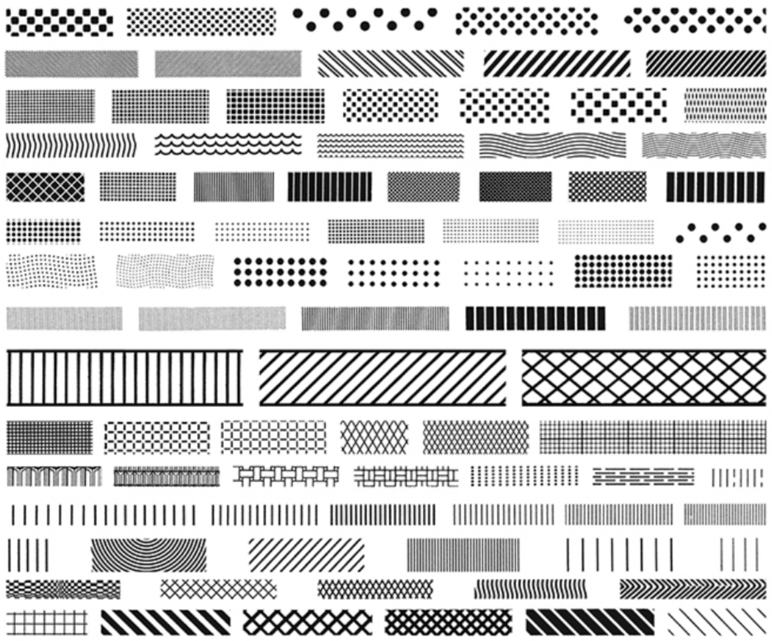
Tufte advocates for removing certain things that people would normally include:

- Grid lines
- Hatching
- Needless ornamentation

Sometimes it gets a bit radical…

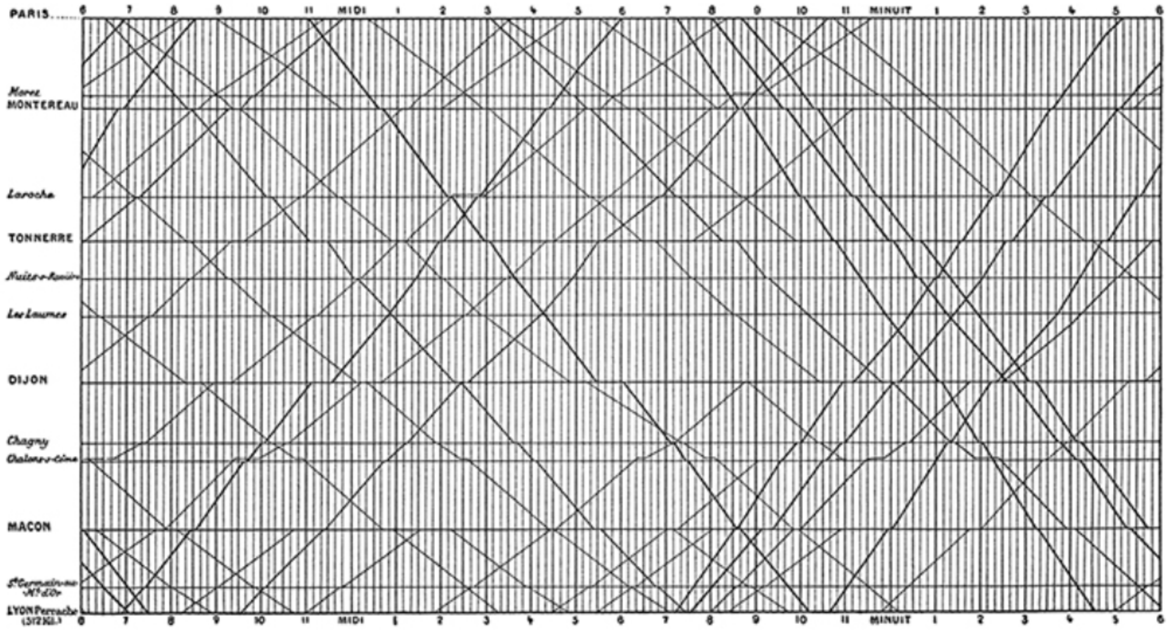






Erasing “chart junk”

The grid in the classic Marey train schedule is very active:



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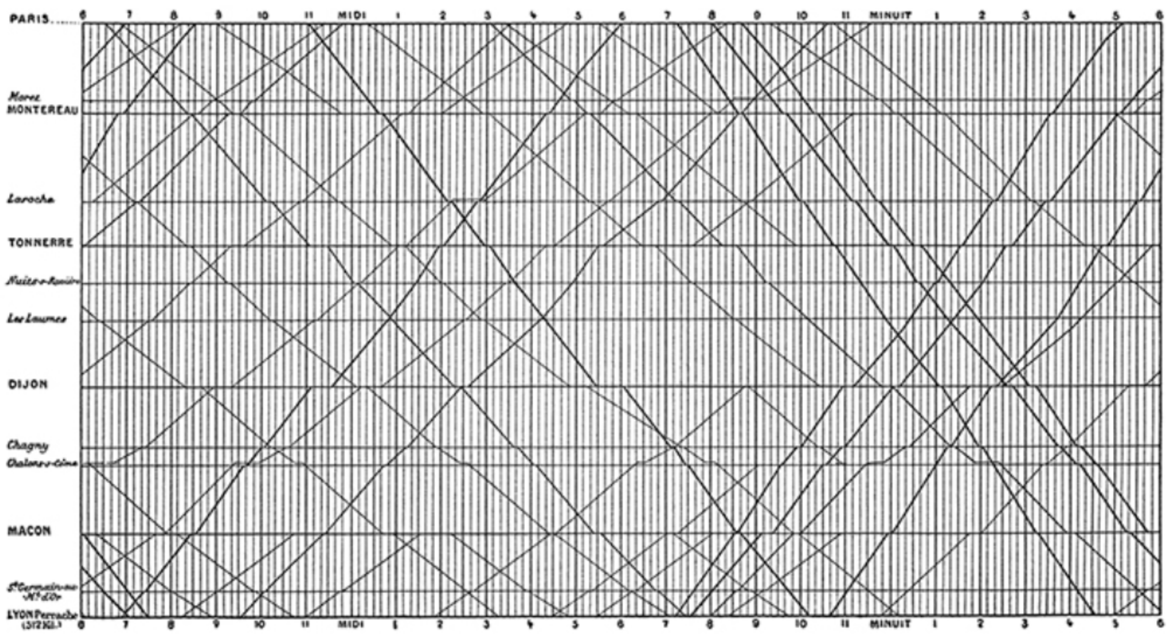
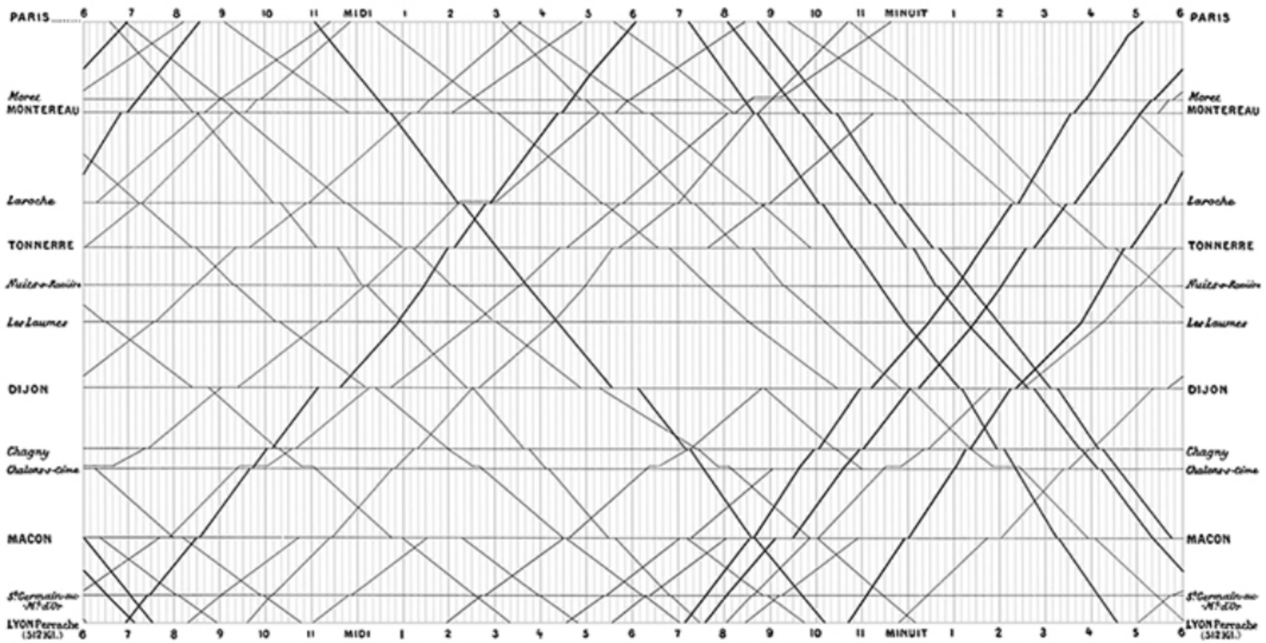


A better treatment, however, is a *gray grid*:

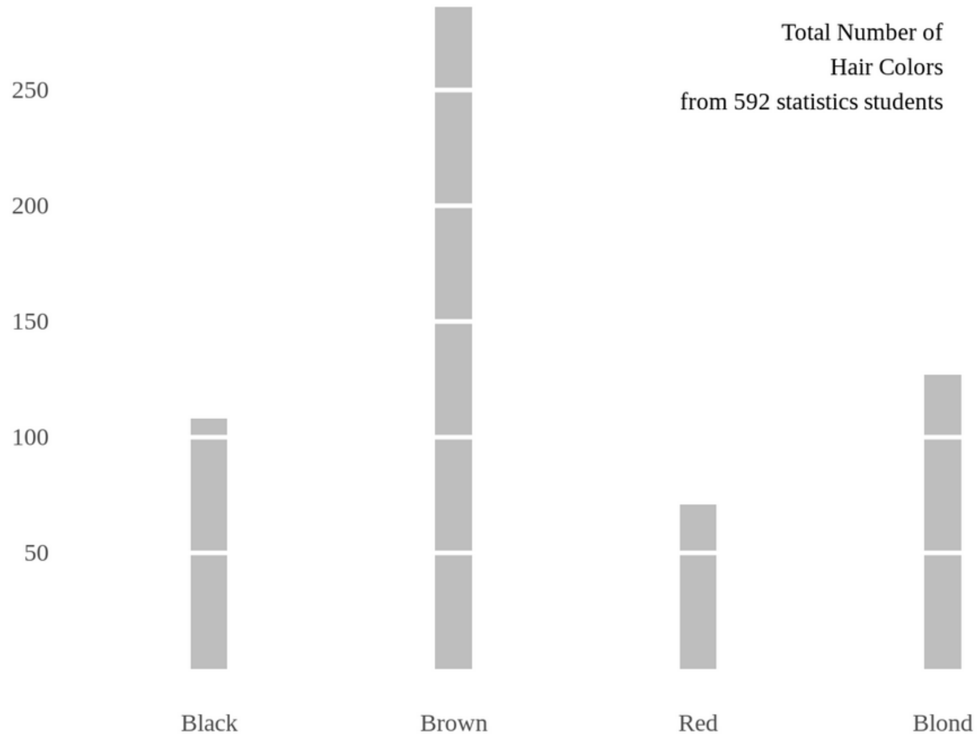


French train schedule looks better without strong gridlines!

Erasing redundant ink (within reason)

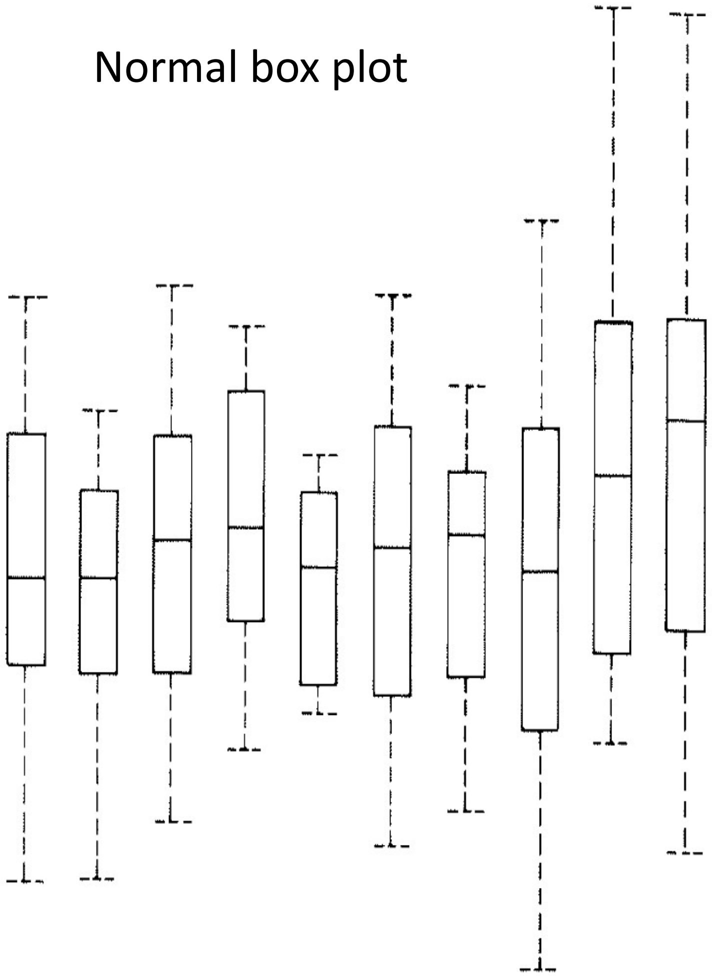
Tufte advocates going further than most people would be comfortable with, erasing things like:

- Parts of the axis not in the data
- Axes that are not being used
- Anything not strictly necessary to show the data.



Erasing redundant ink (within reason)

Normal box plot



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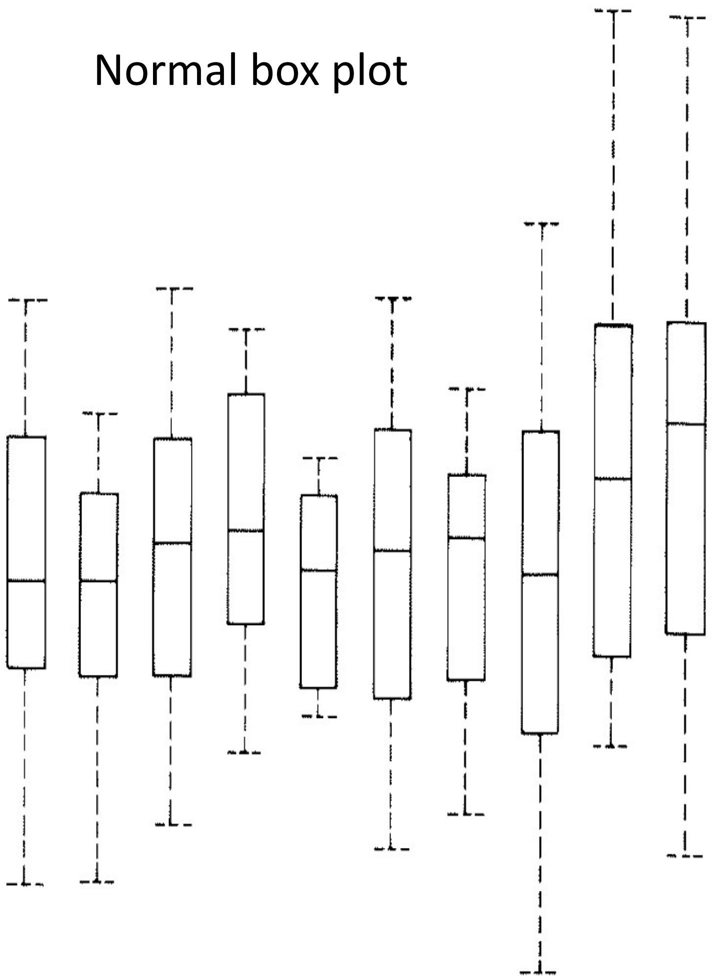
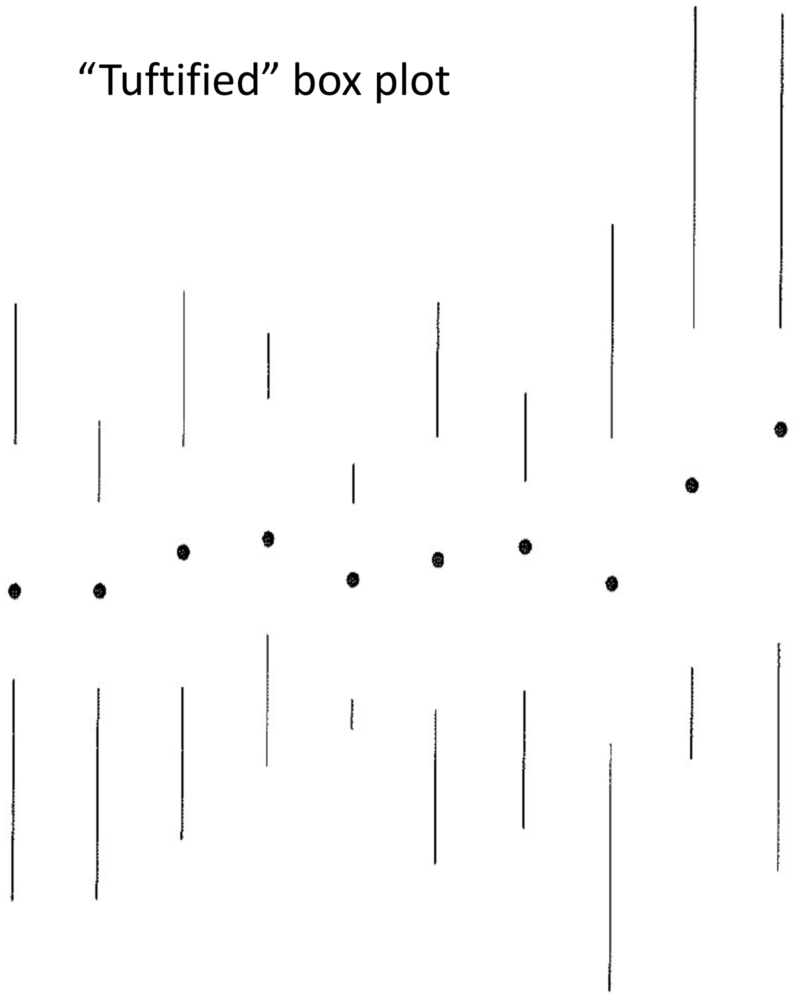
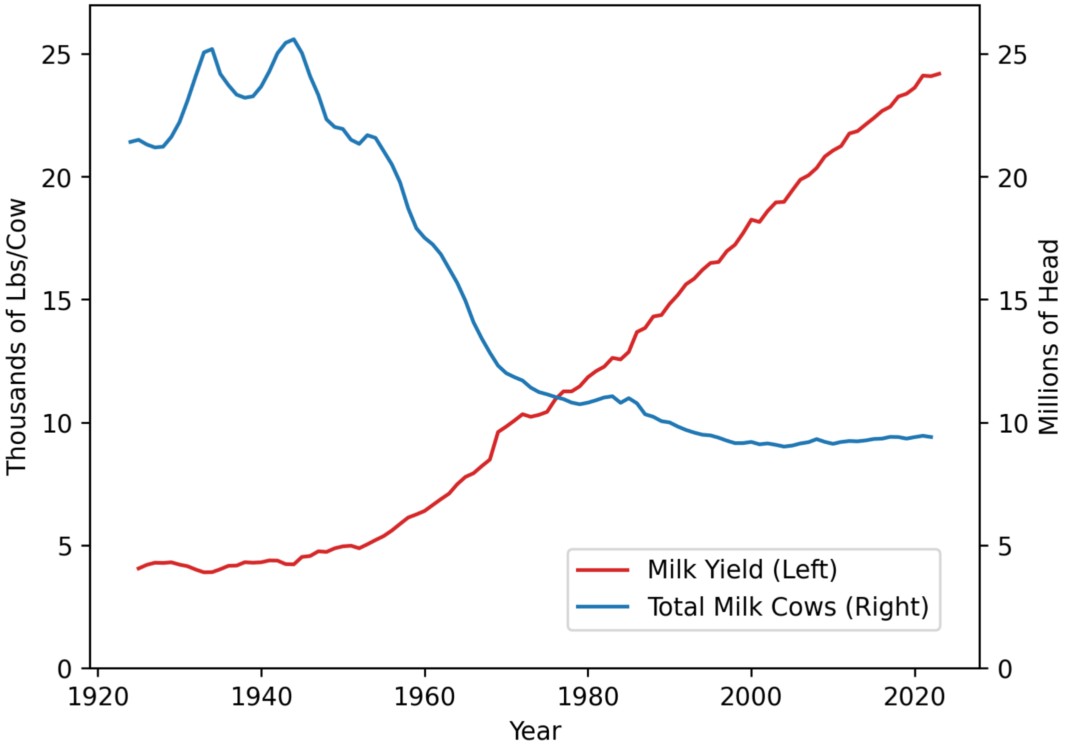


“Tuftified” box plot



Erasing redundant ink (within reason)

My first attempt



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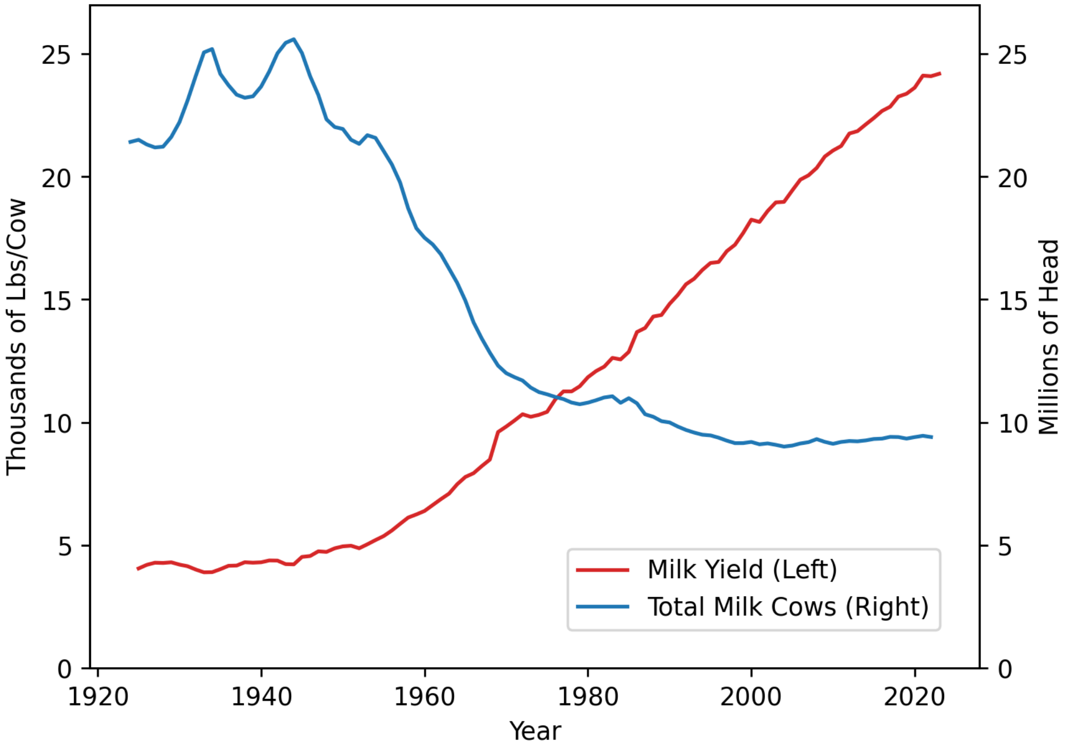
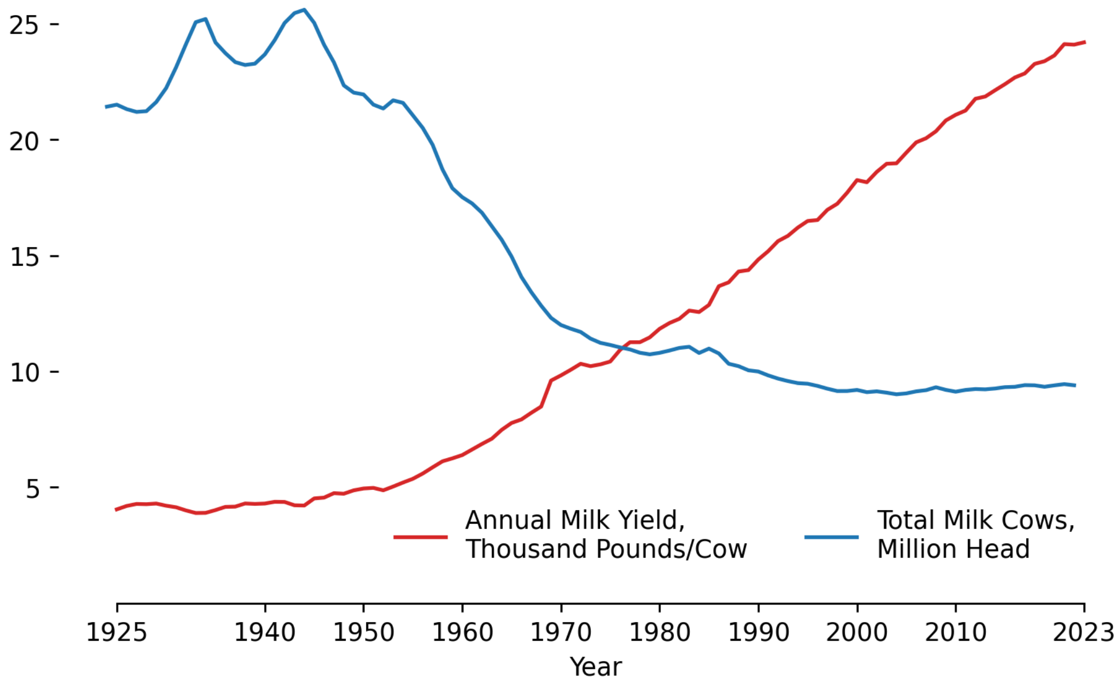
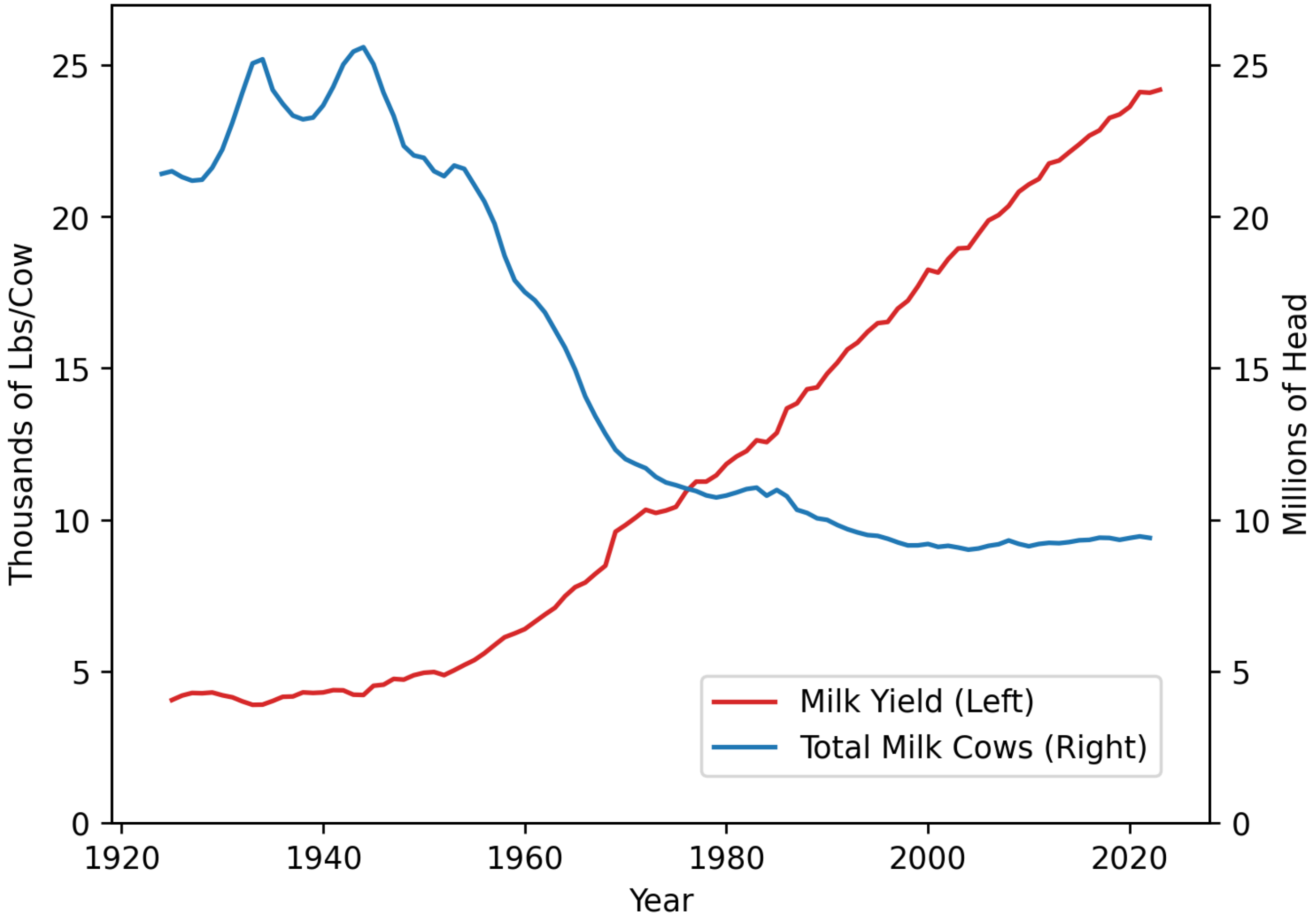


Applying Tufte edits



Summarizing some of my Tufte edits:



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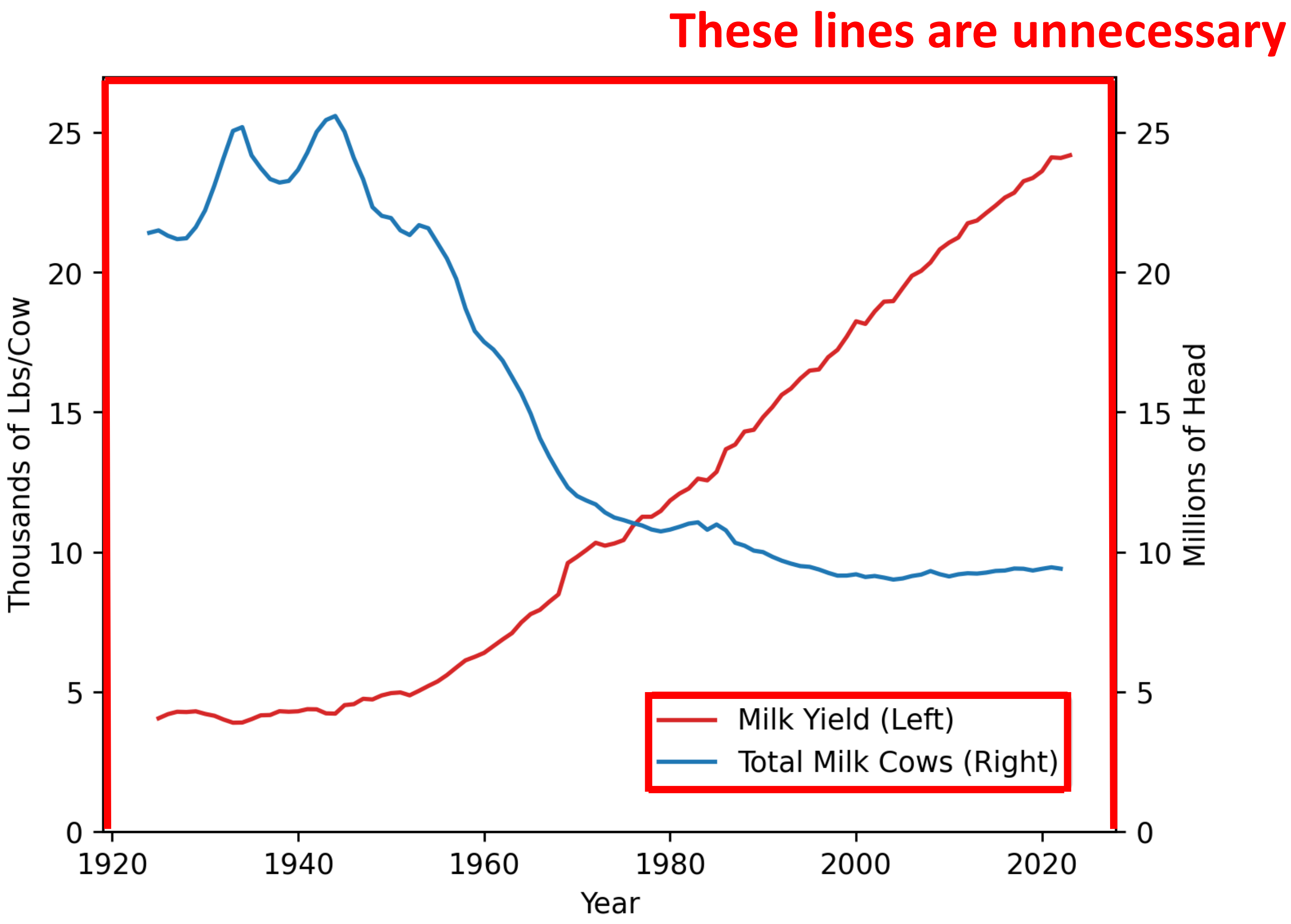
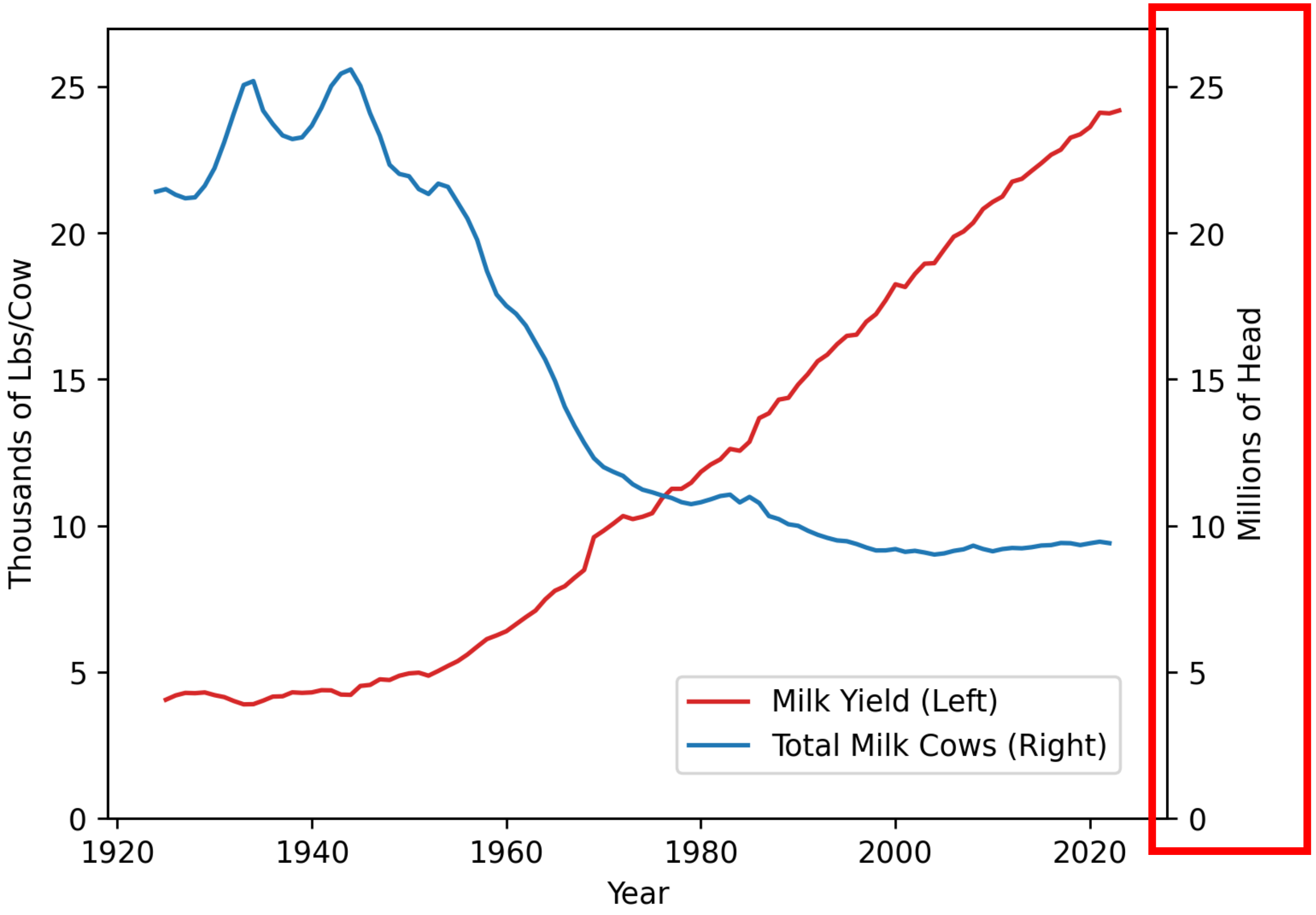


Summarizing some of my Tufte edits:

This scale is the same as the left one



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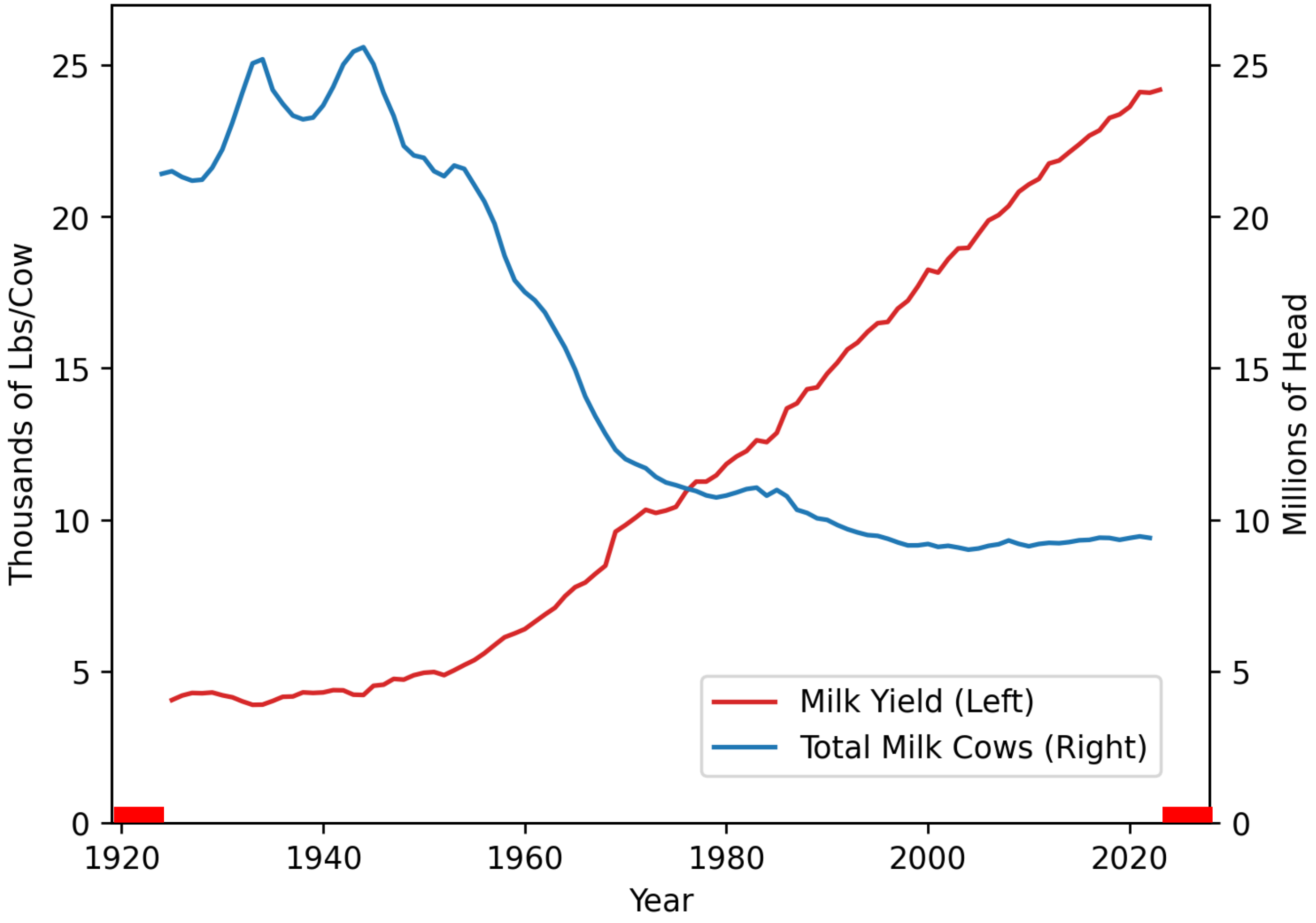
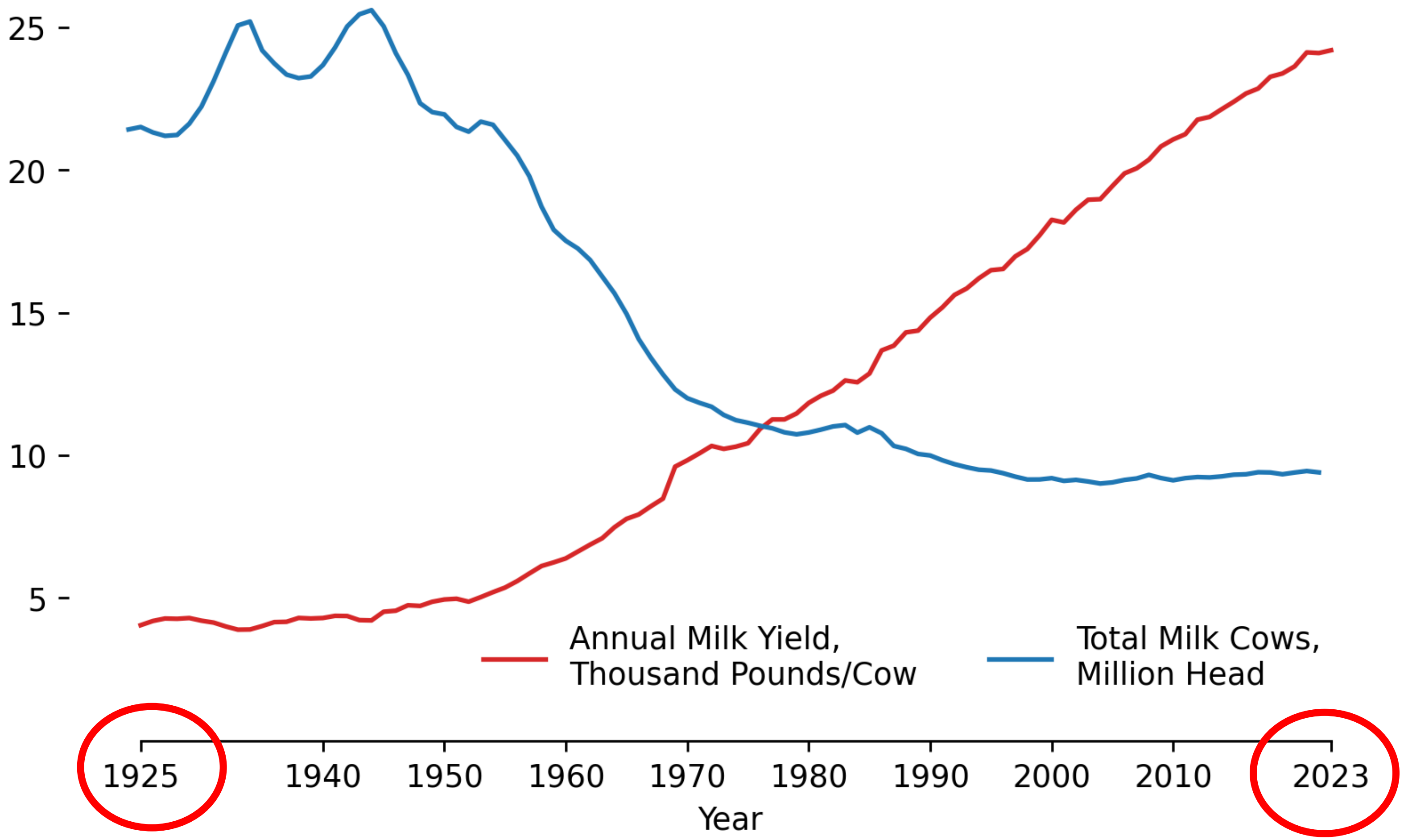


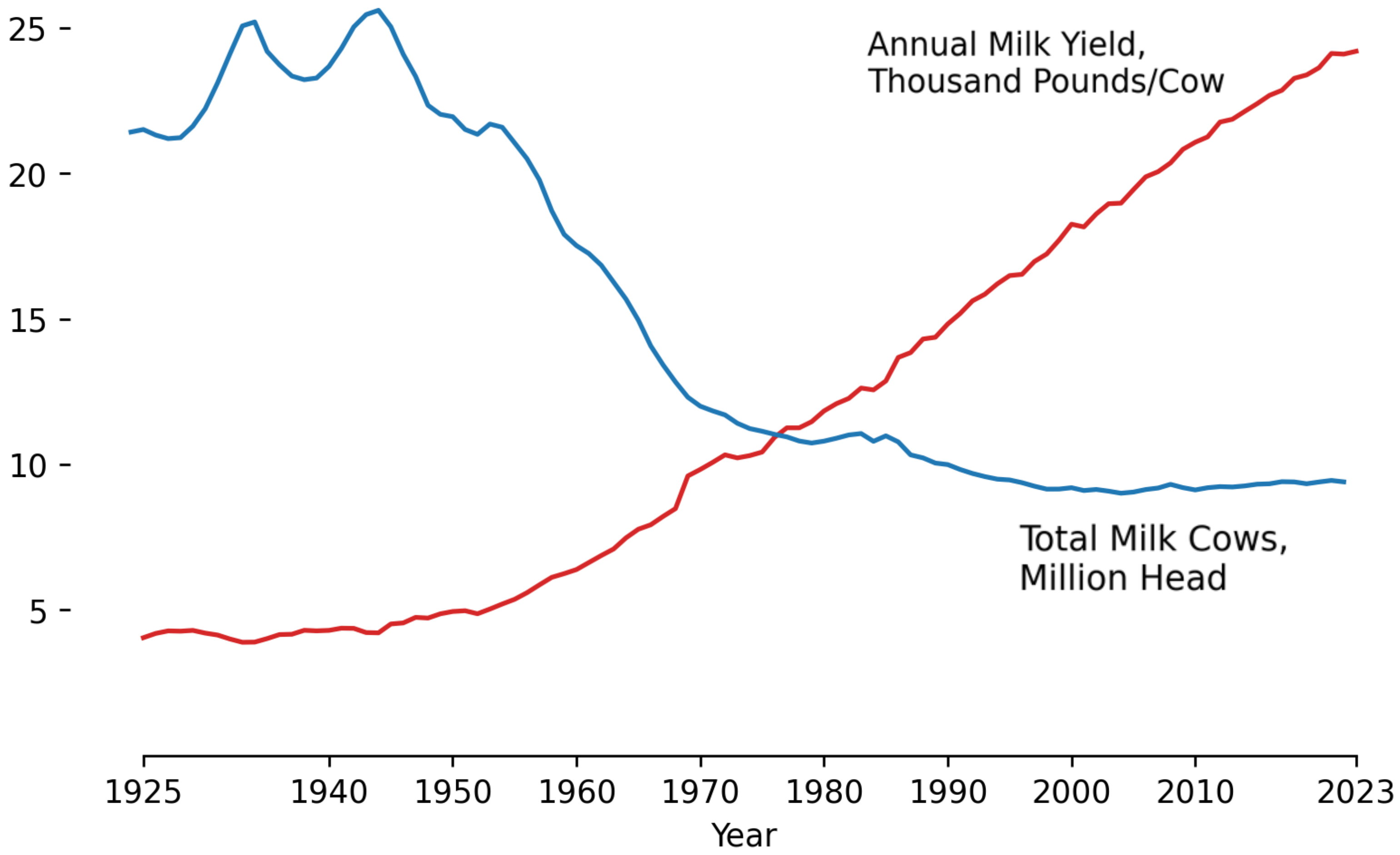
These x points are not actually in the data

Summarizing some of my Tufte edits:



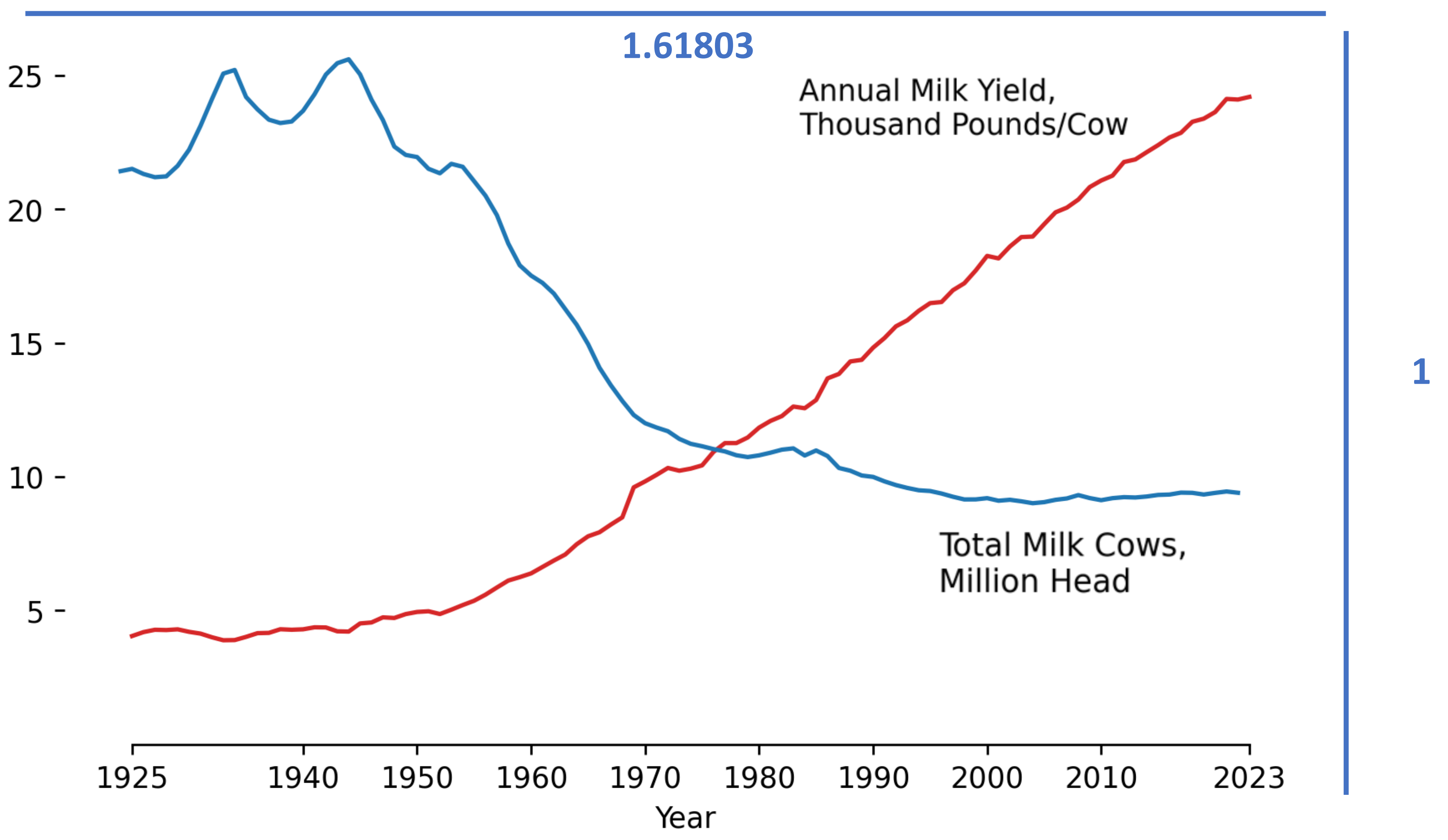
I can then change the end points to show where the data begin and end!

Summarizing some of my Tufte edits:



Bonus edit: can remove the lines from the legend entirely

Summarizing some of my Tufte edits:

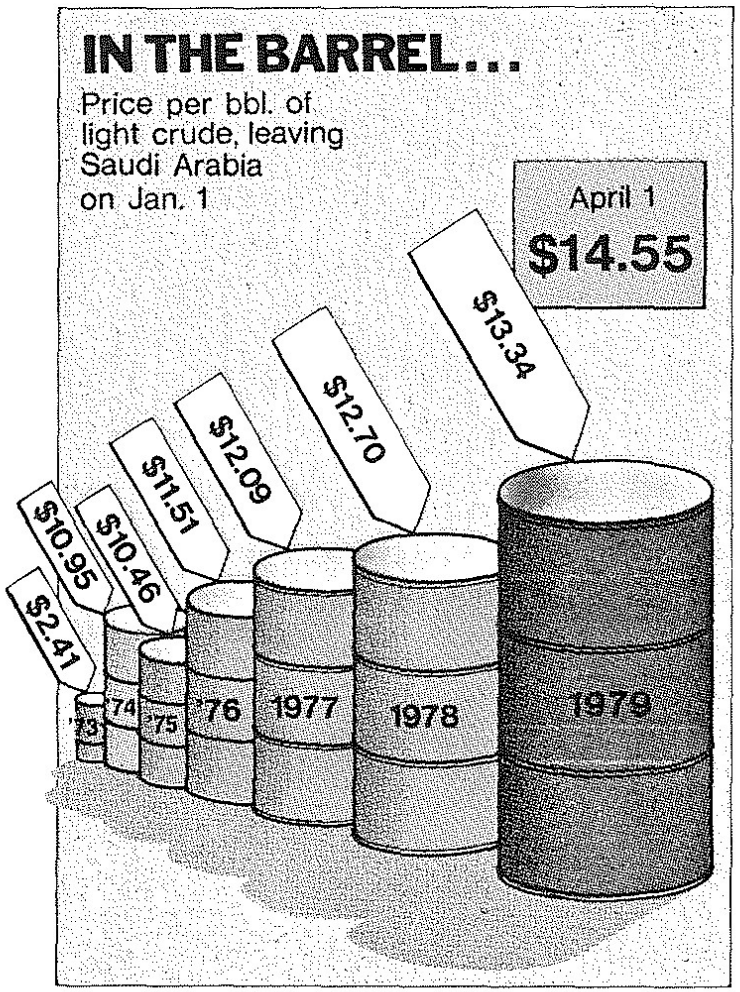


My own flavor: I like using the golden ratio for the size of figures.

Proportional ink principle

"The representation of numbers, as physically measured on the surface of the graphic itself, should be directly proportional to the numerical quantities represented"

If twice the ink, should be twice the value!

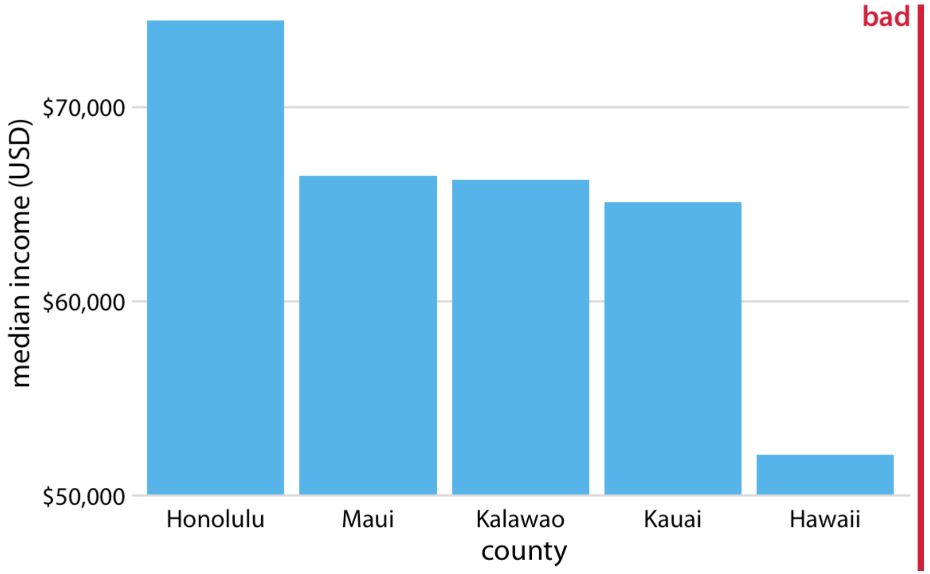


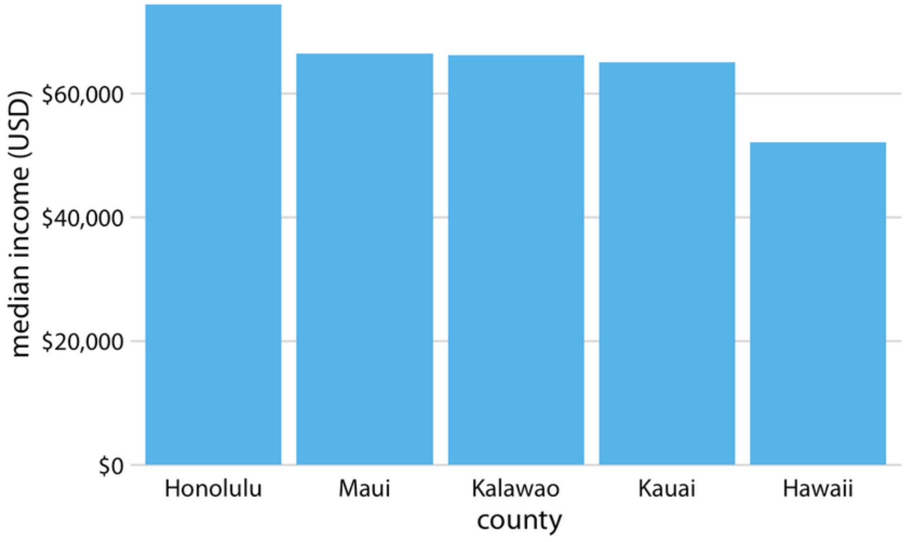
Proportional ink principle

Common ways to violate this principle:

1. Truncating the y-axis

[Example](#)

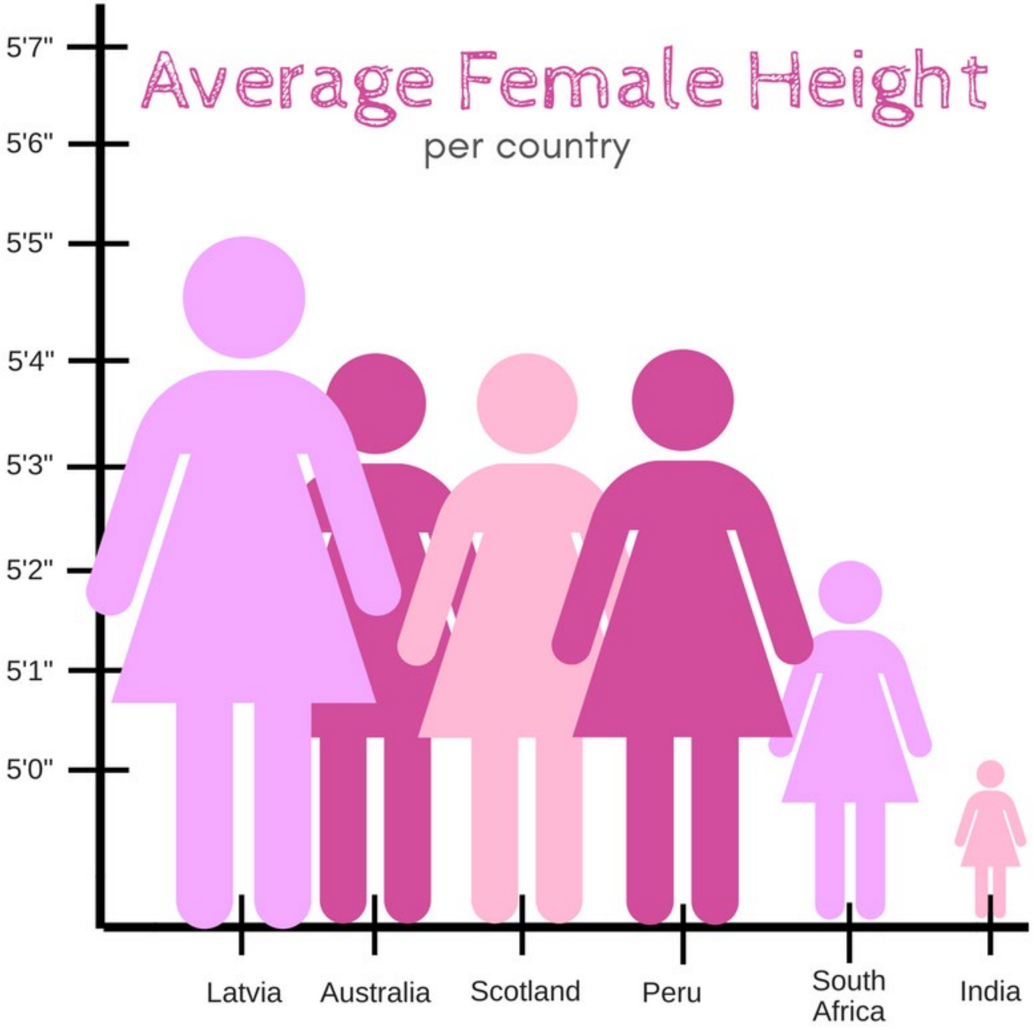




Proportional ink principle

Common ways to violate this principle:

1. Truncating the y-axis
2. Univariate data portrayed with multiple dimensions

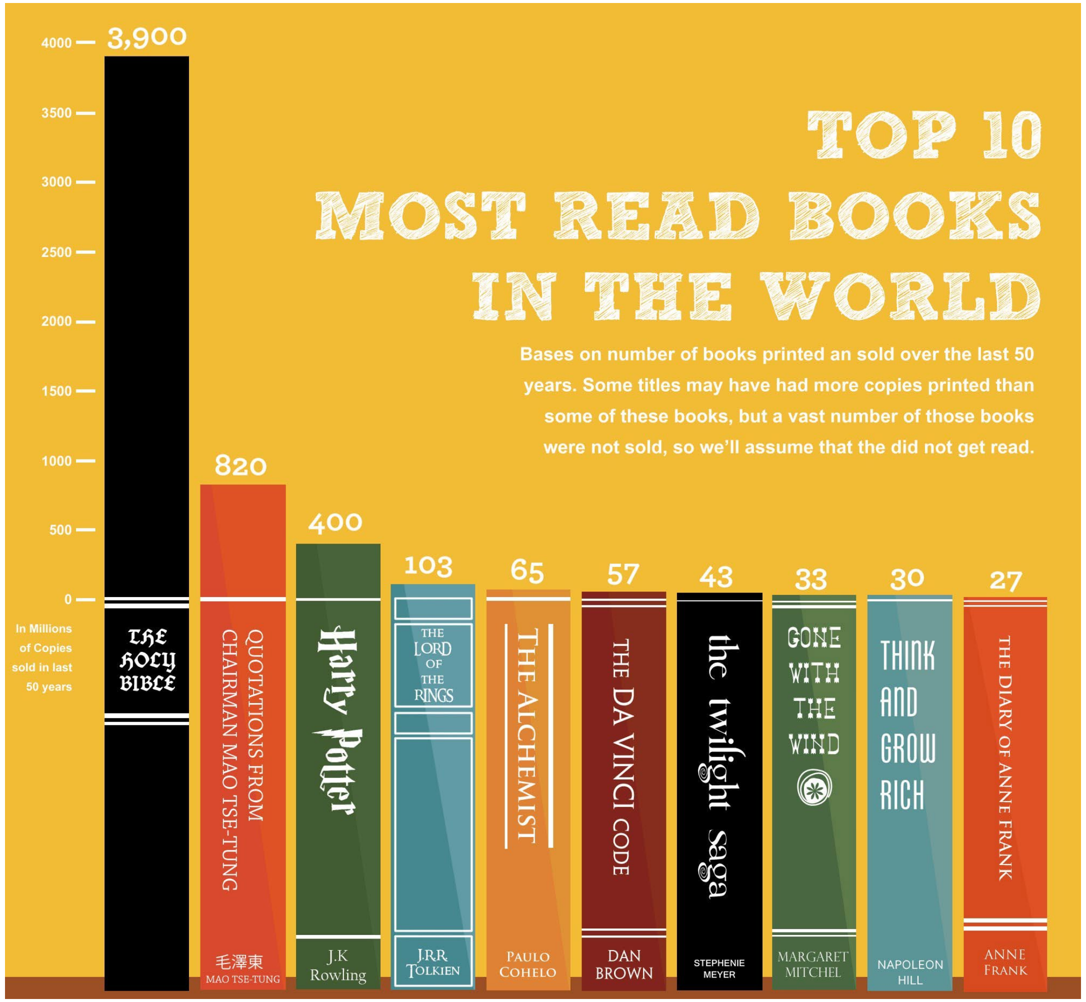


Proportional ink principle

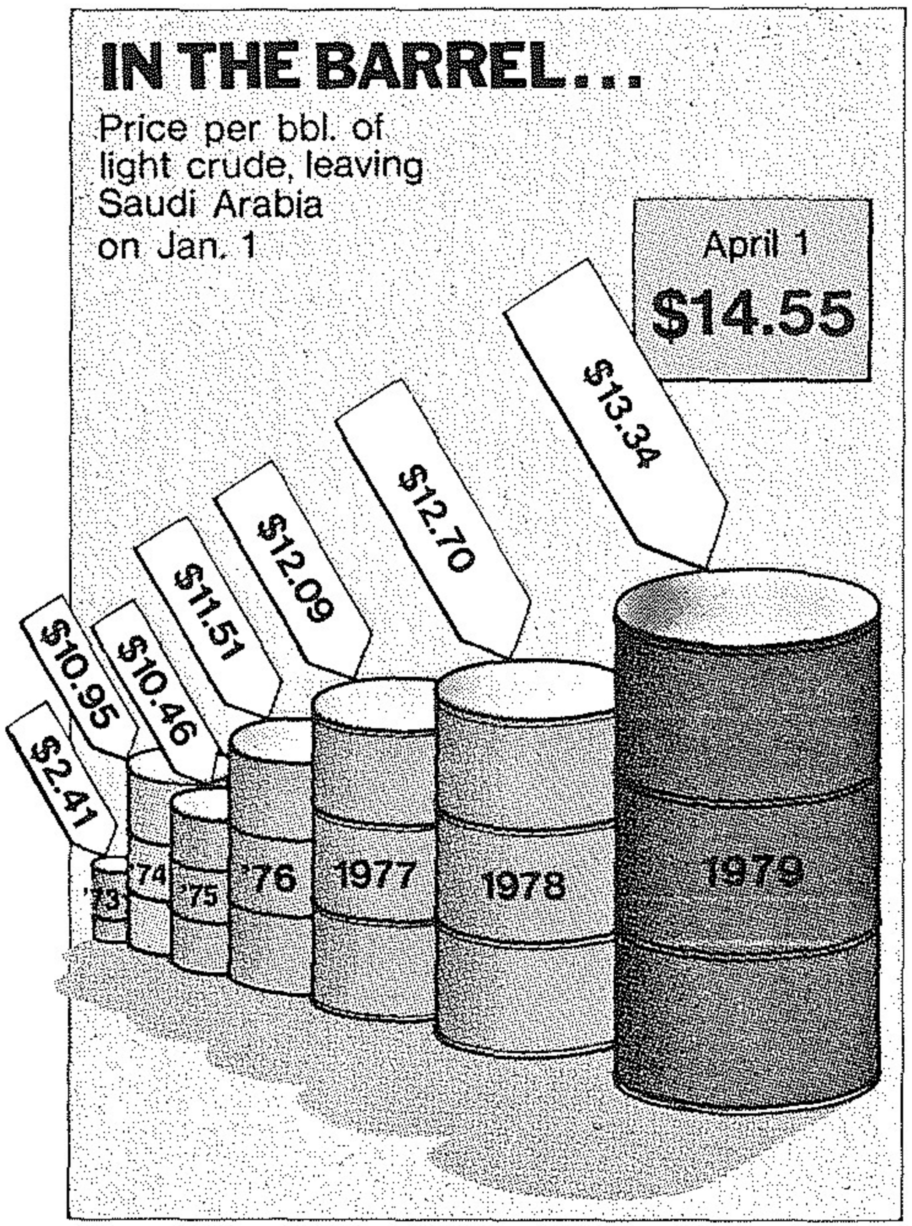
[Link]

Common ways to violate this principle:

1. Truncating the y-axis
2. Univariate data portrayed with multiple dimensions
3. Design elements



Where is the violation here?



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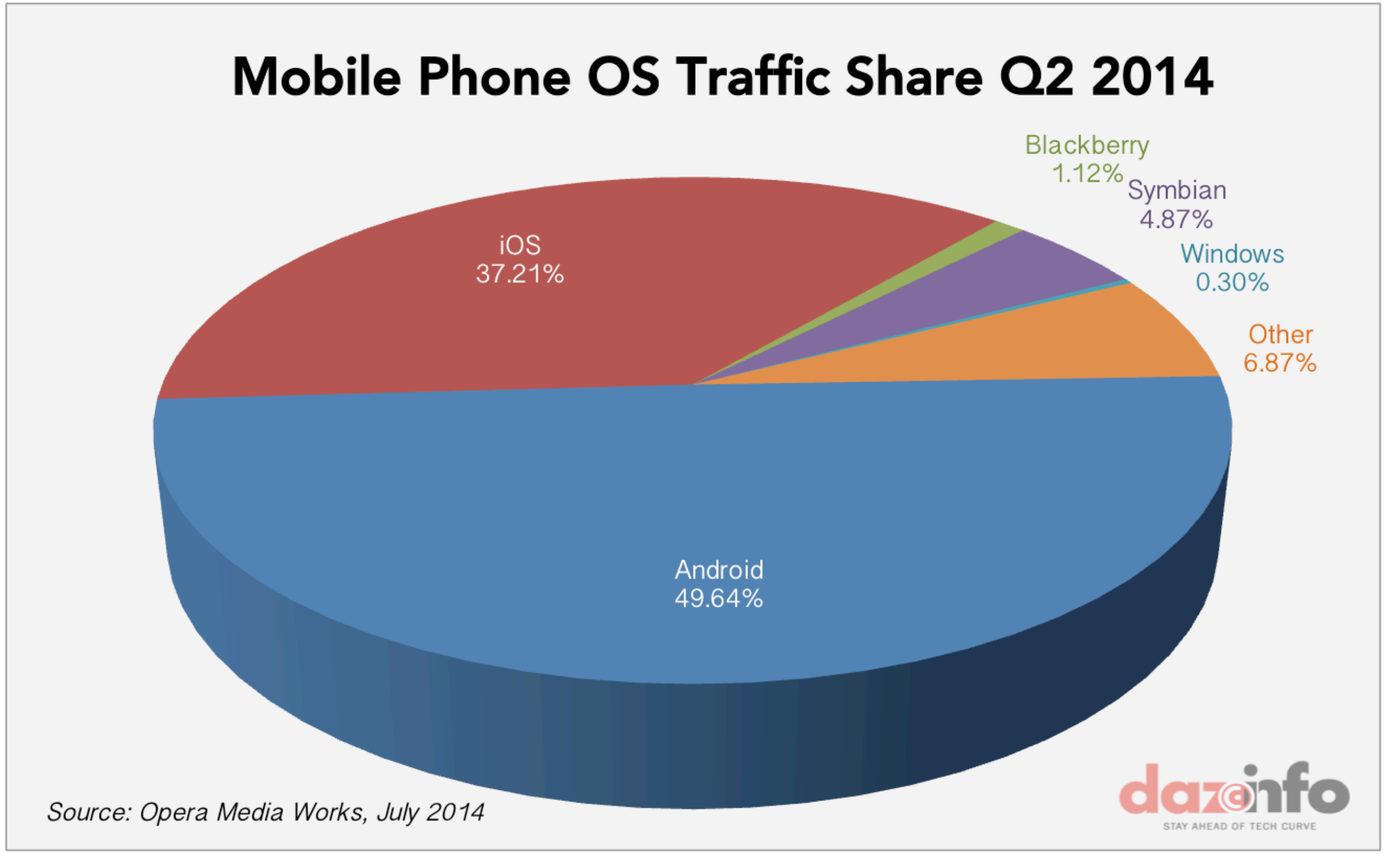

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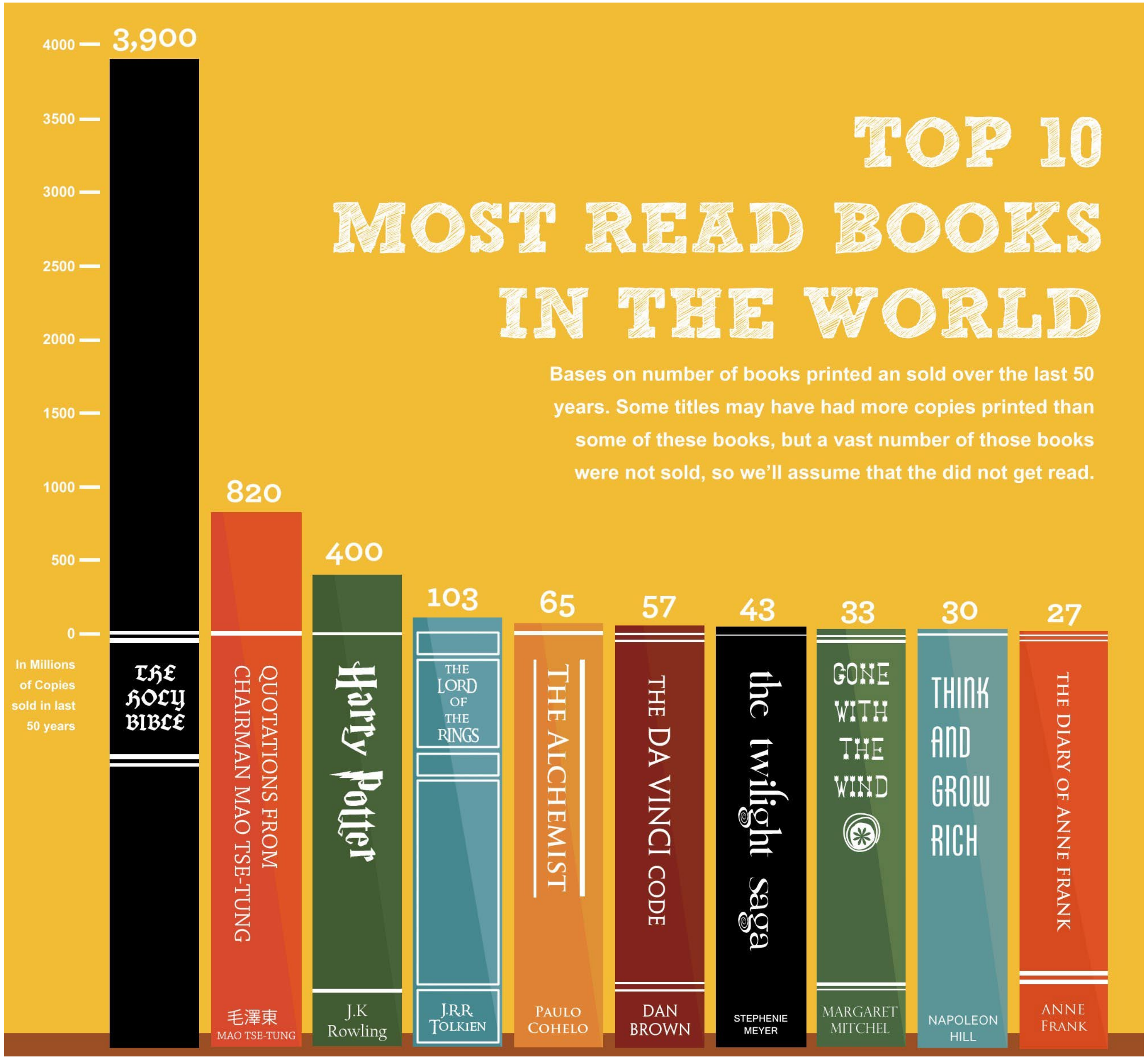
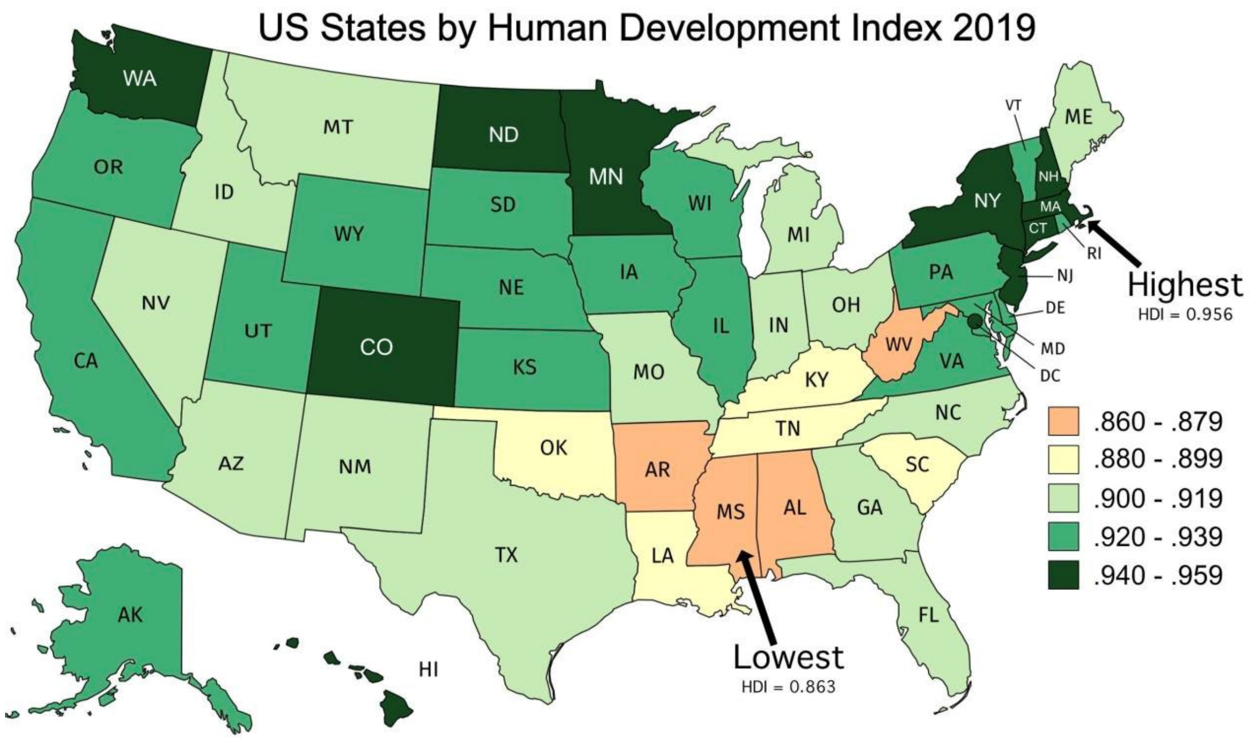


Another principle: show data in context



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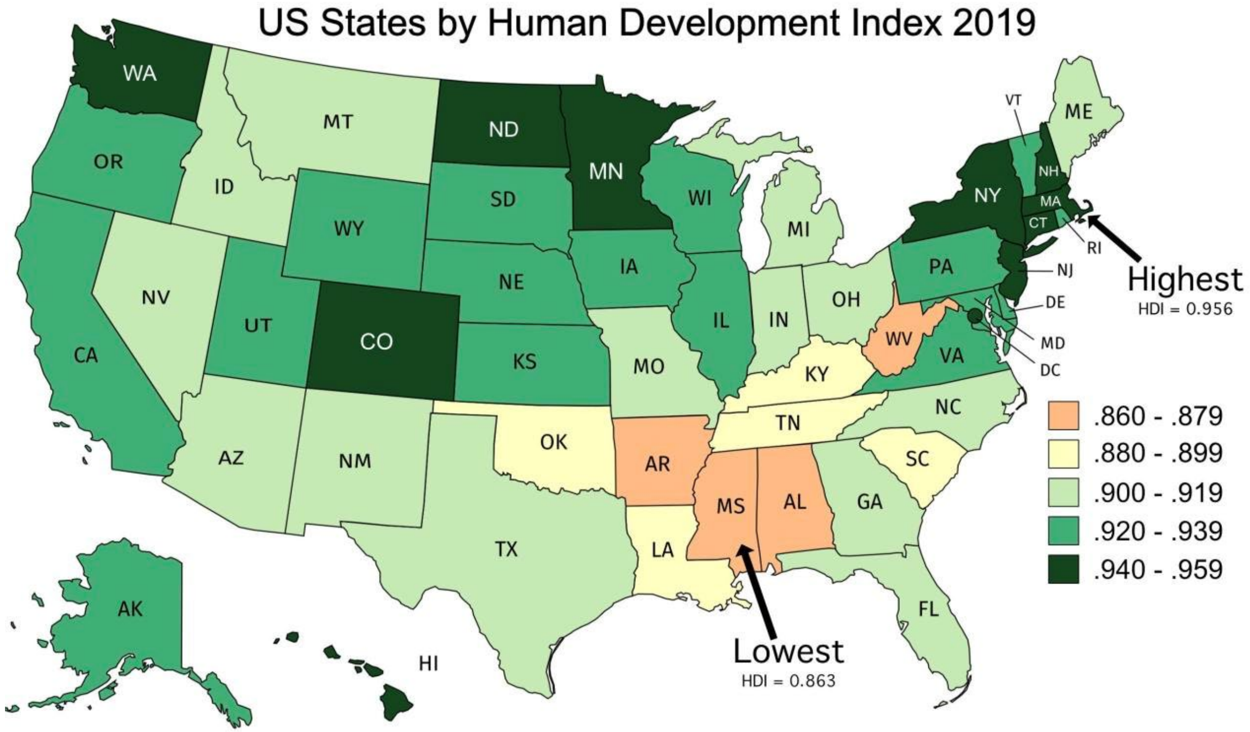
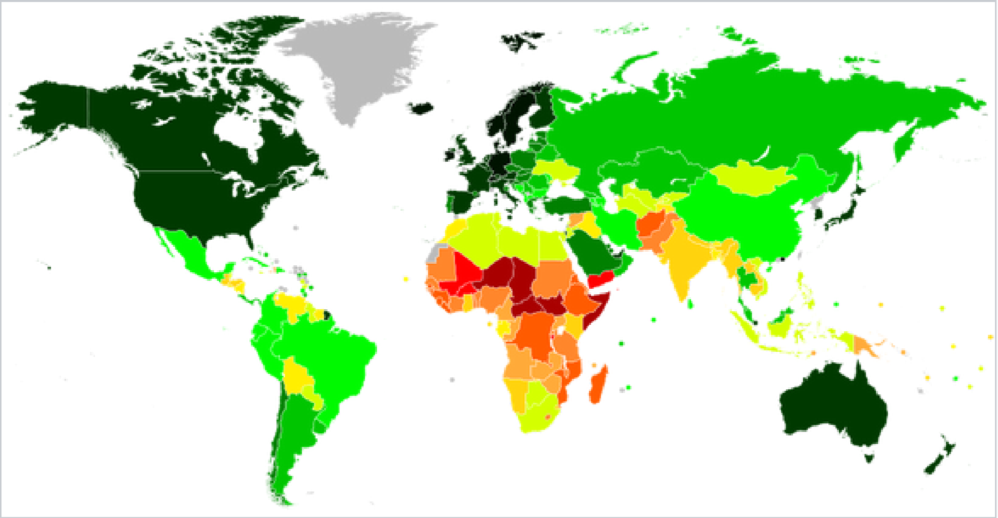




World map of countries and territories by HDI scores in increments of 0.050 (based on 2022 data, published in 2024)

- ≥ 0.950
- 0.900–0.950
- 0.850–0.899
- 0.800–0.849
- 0.750–0.799
- 0.700–0.749
- 0.650–0.699
- 0.600–0.649
- 0.550–0.599
- 0.500–0.549
- 0.450–0.499
- 0.400–0.449
- ≤ 0.399
- Data unavailable

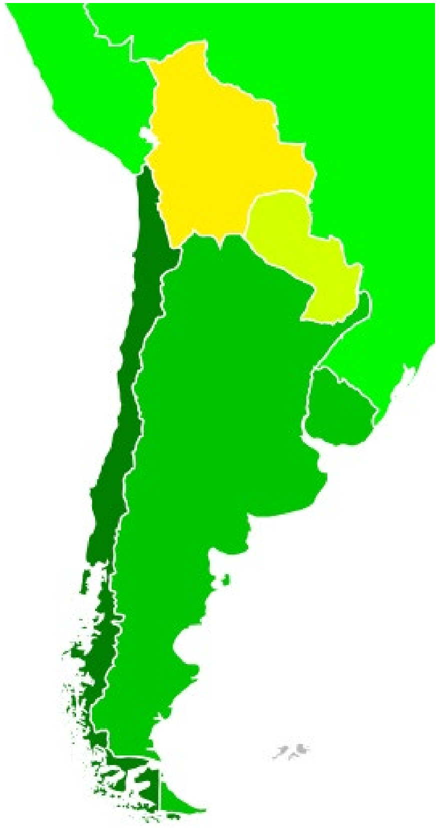
Another principle: show data in context

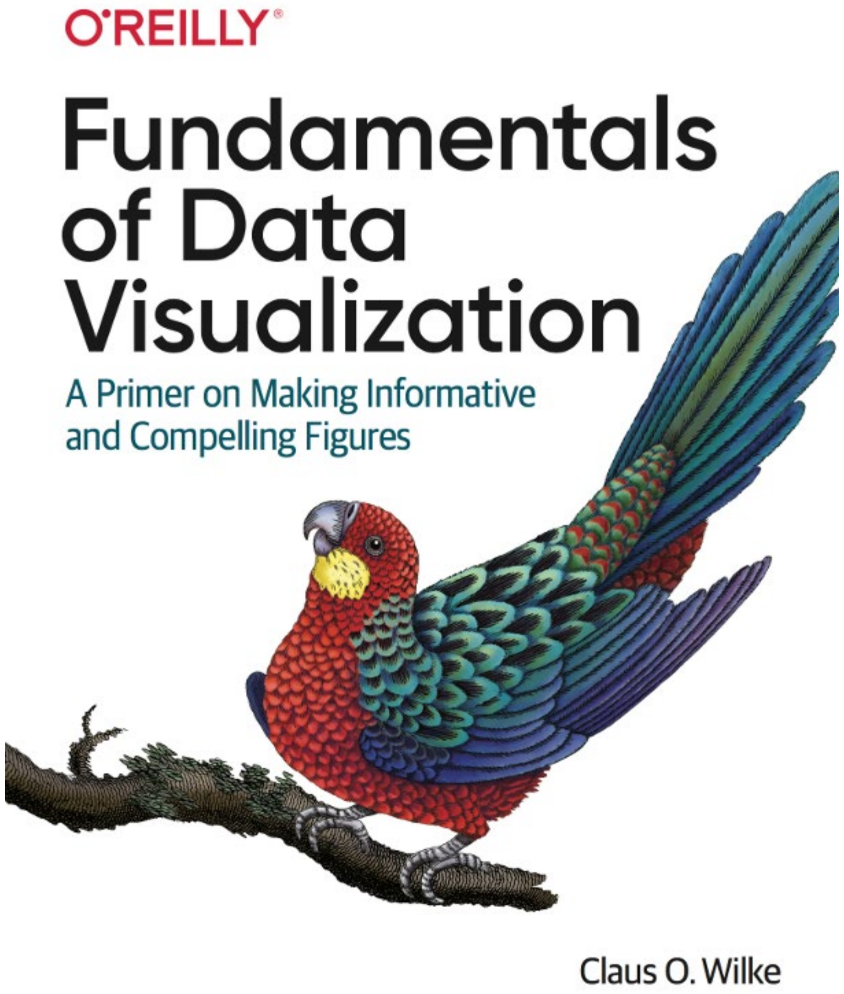
Tufte points to one way to deceive people, which is to not give proper context to data.

In the HDI map of the US, four states are singled out to be red, like in the country graph…

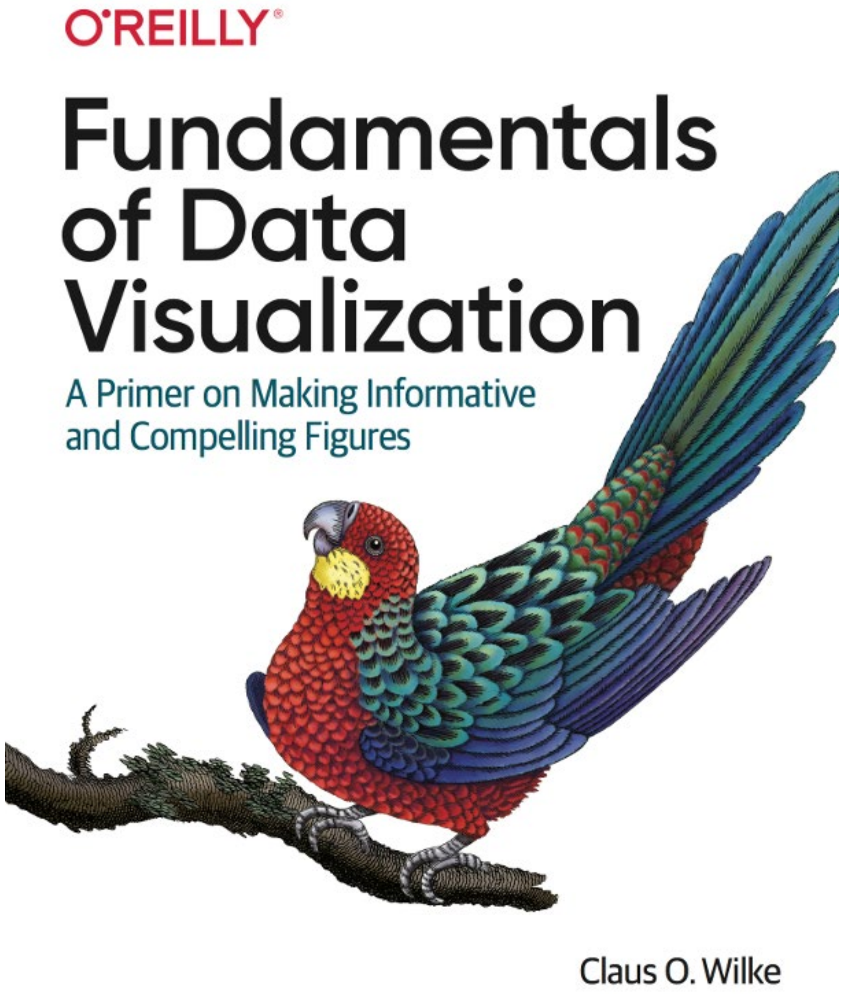
… even though they are really somewhere between Chile and San Marino, dark green on the map.

San Marino	0.867
Chile	0.860





Data Visualization Principles: Lessons on Color from Wilke



Claus Wilke's book is a free text that has some particularly good guidance about using color.

Wilke mentions three ways to use color:

1. Color as a tool to distinguish
2. Color to represent data values
3. Color as a tool to highlight

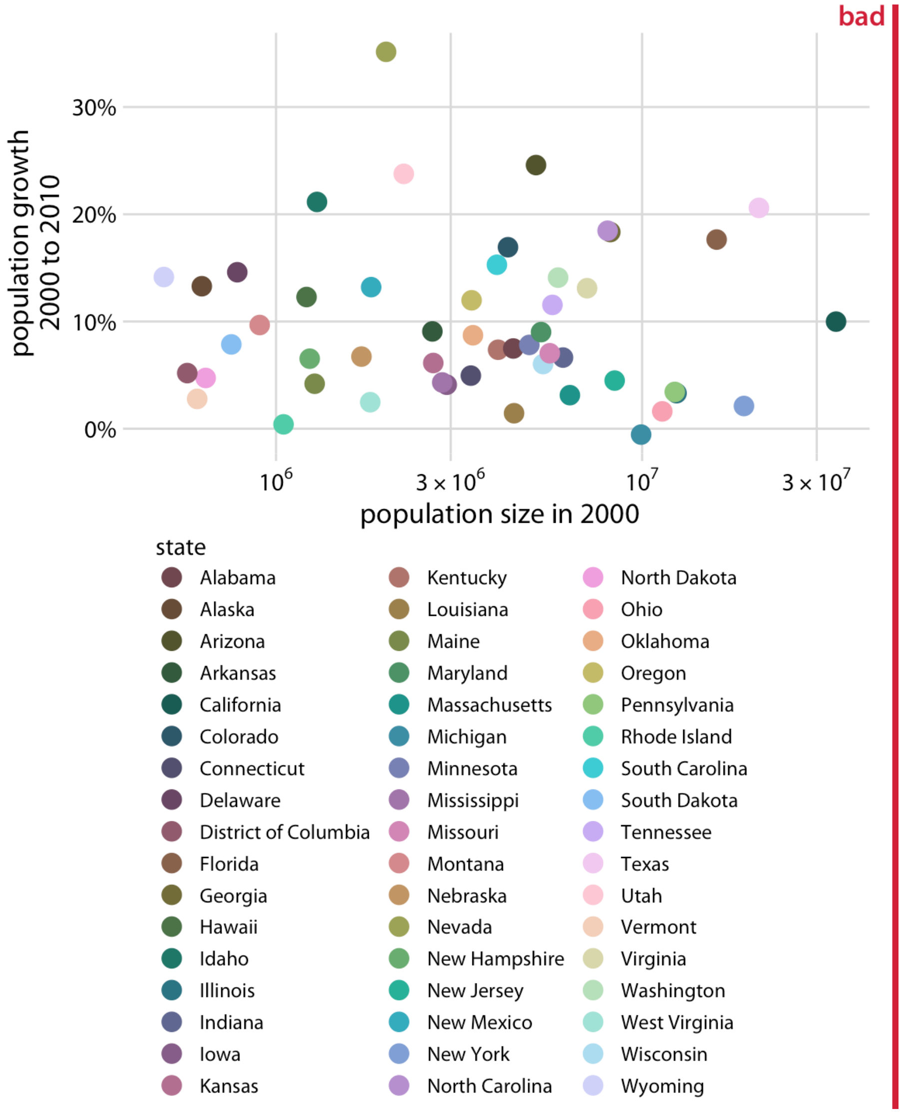
I will add: Color in context

Color to distinguish

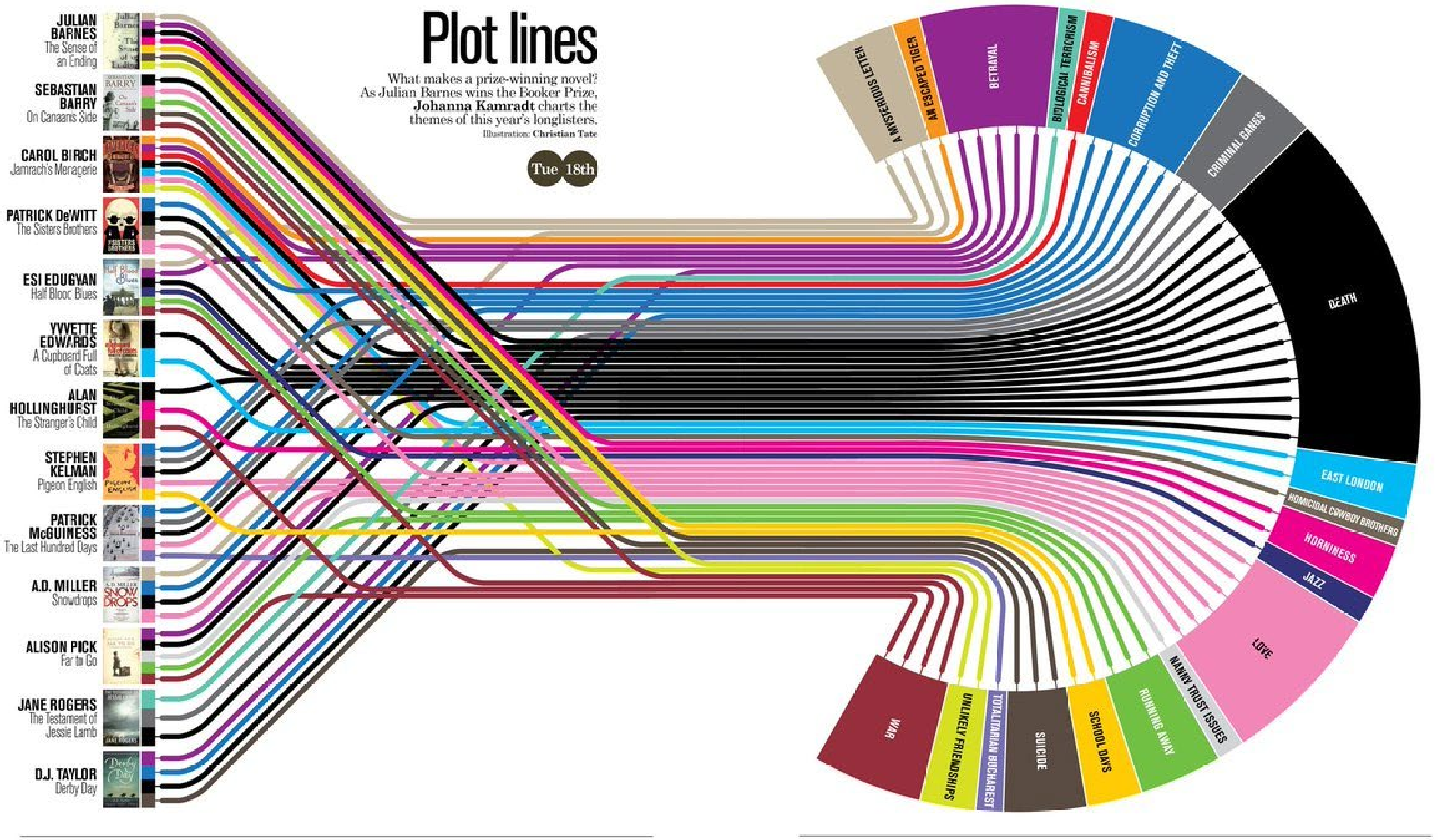
Color used this way is for discrete values and categories, not continuous variables.

Rule of thumb: people ***cannot distinguish more than 5 categories*** with color.

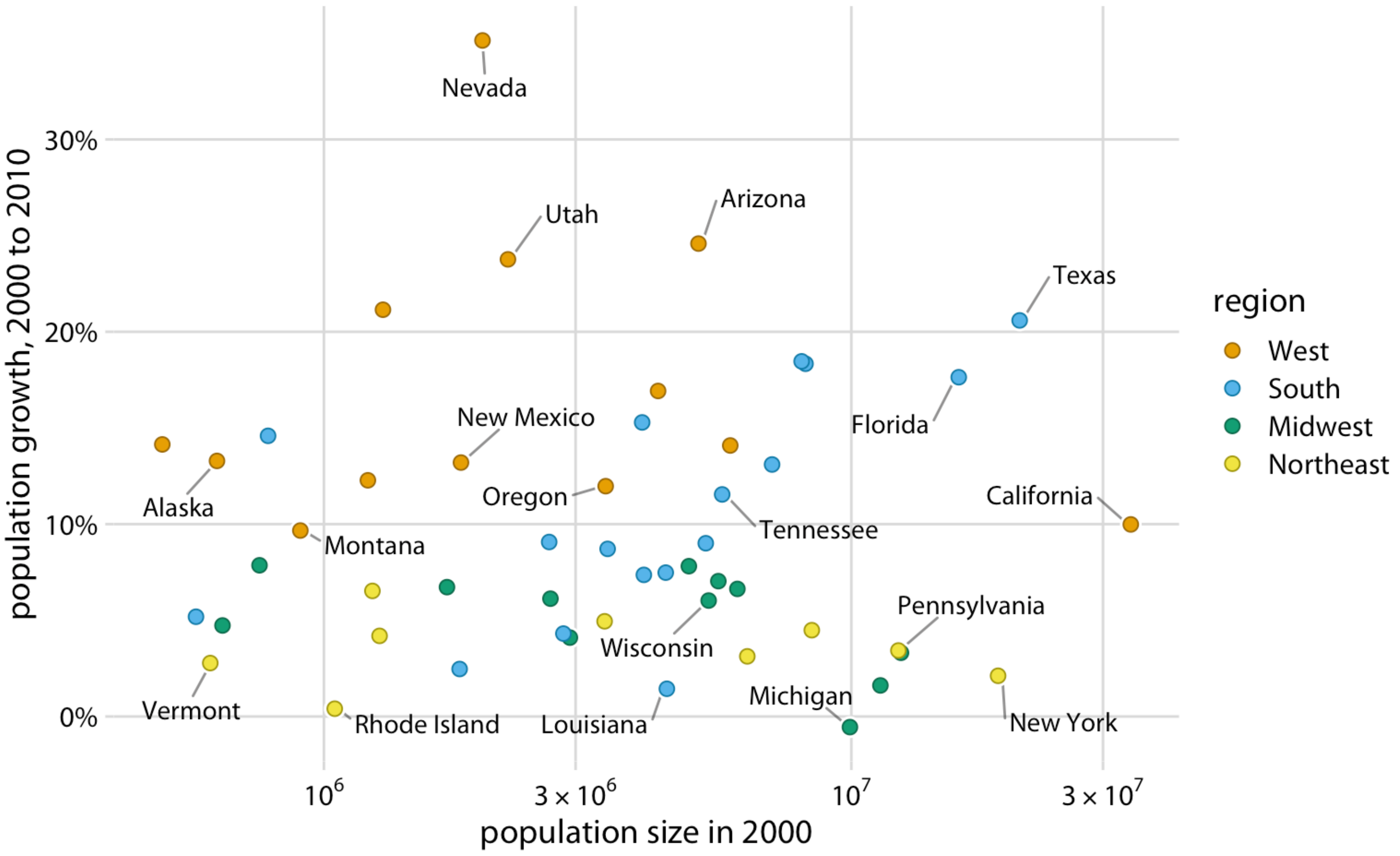
(I see a shocking number of graphs at the PhD level that can't get this right).



Color to distinguish



Color to distinguish

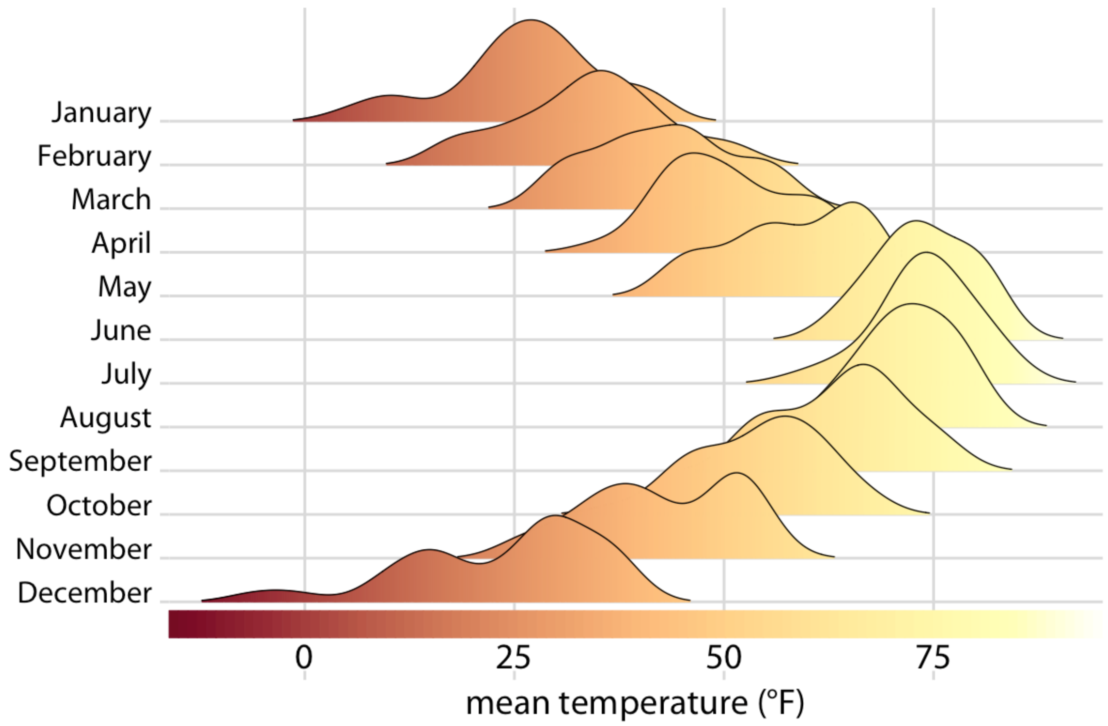


Color to represent data values

To represent a continuous variable, most visualizations use a colormap.

If the variable is ***continuous,*** the colormap must be ***monotonic.***

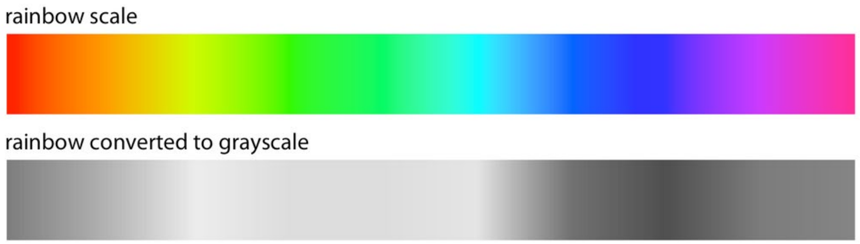
In other words, changes in shade have to be uniform if changes are uniform.

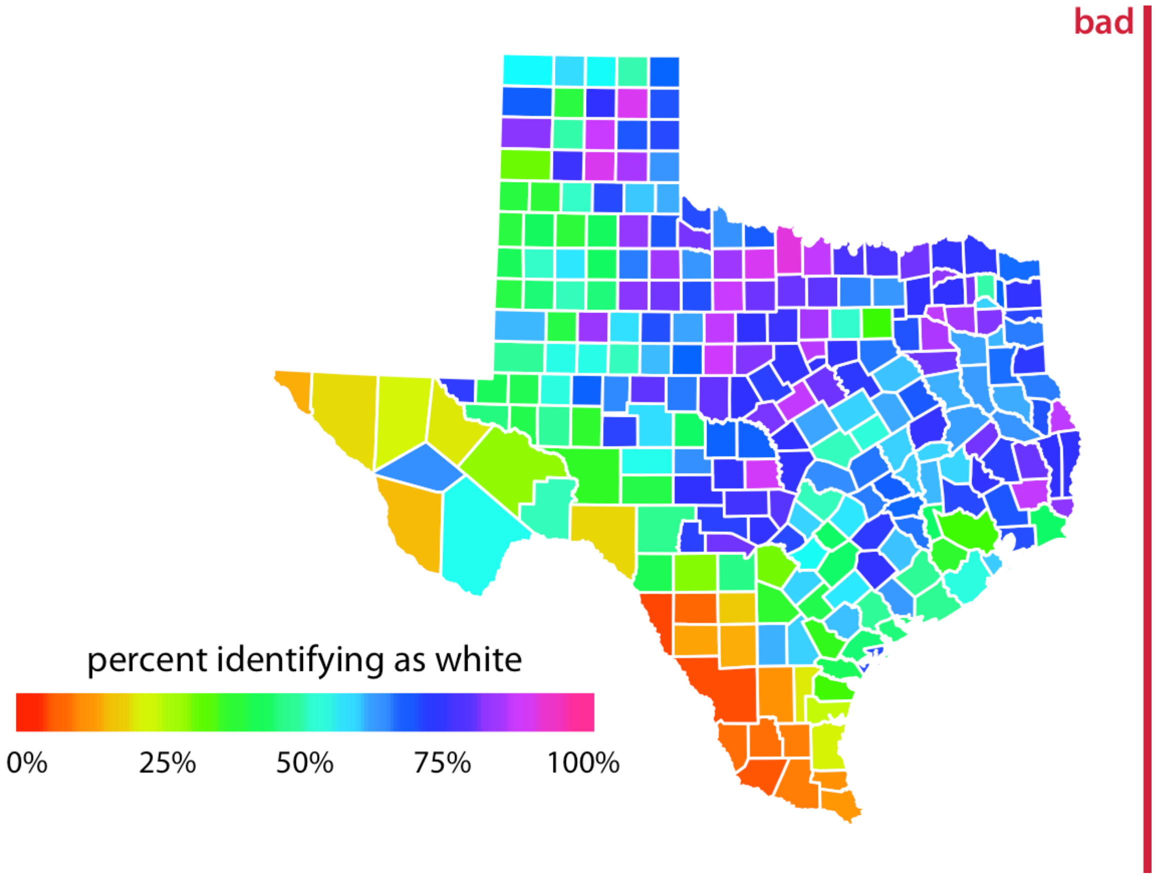


Color to represent data values

The rainbow colormap (often called "Jet") is the most famous culprit.

The changes in the rainbow when mapped to gray clearly don't change consistently with the variable.

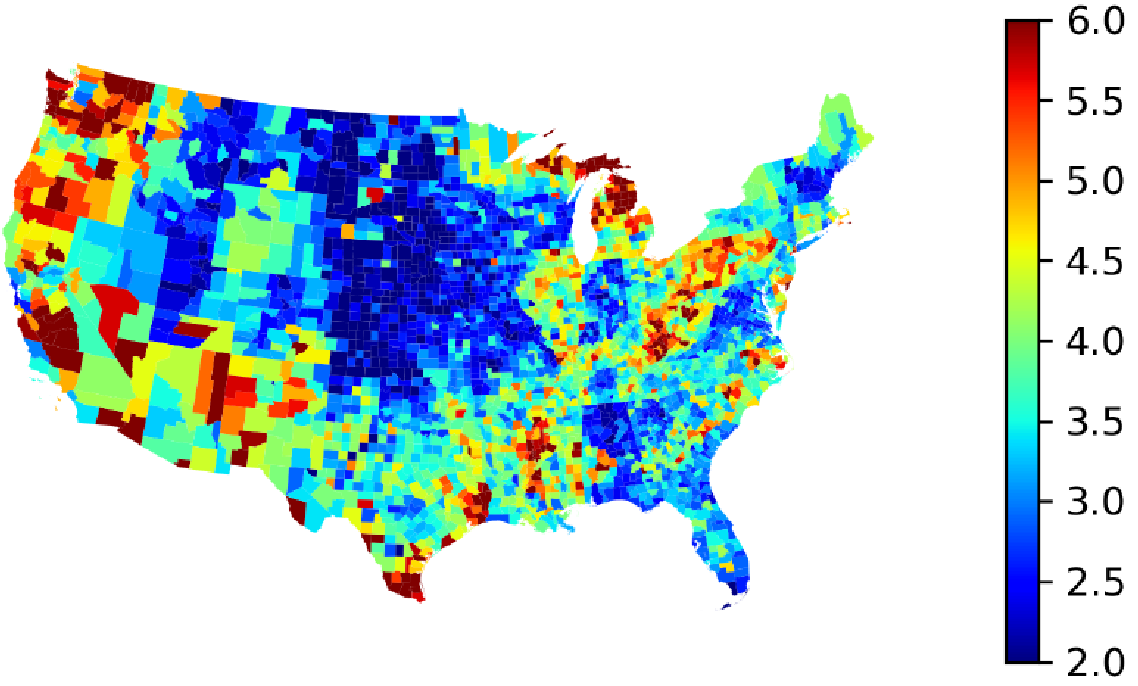




Color to represent data values

What categories are your eyes making when you look at this colormap?

Figure 5: County Unemployment Rates in 2022



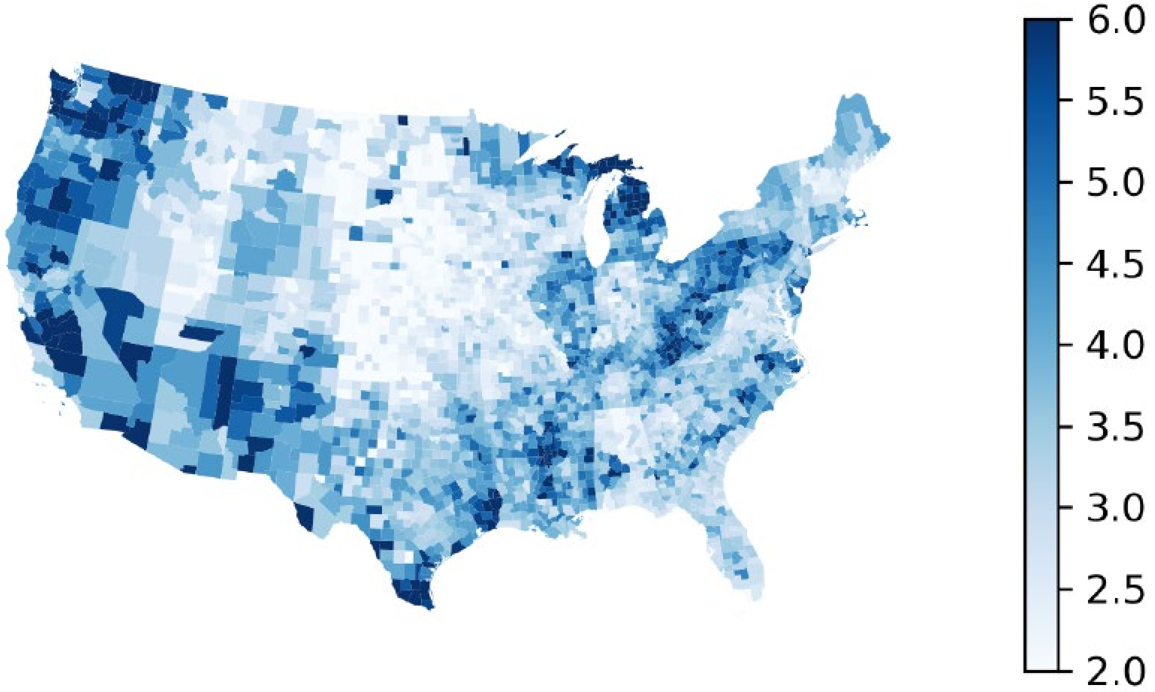
(a) Non-Monotonic Colormap ("Jet")

Color to represent data values

What categories are your eyes making when you look at this colormap?

How about now?

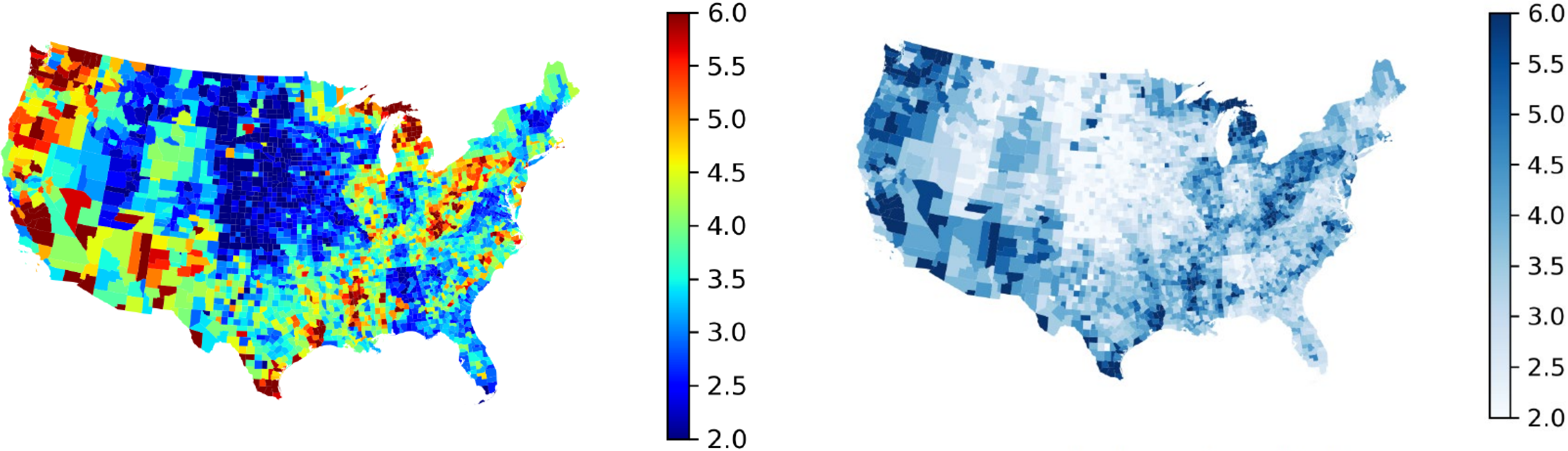
Figure 5: County Unemployment Rates in 2022



(b) Monotonic Colormap ("Blues")

Color to represent data values

Figure 5: *County Unemployment Rates in 2022*



(a) *Non-Monotonic Colormap ("Jet")*

(b) *Monotonic Colormap ("Blues")*

Color to represent data values

Color maps like the ones here are best for discrete categories, not continuous data

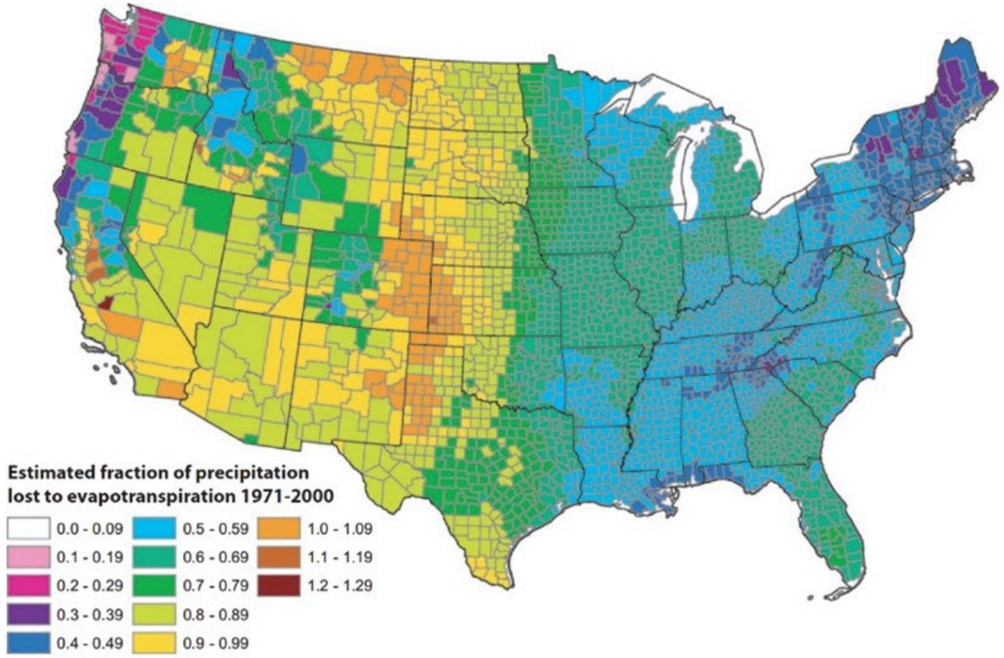


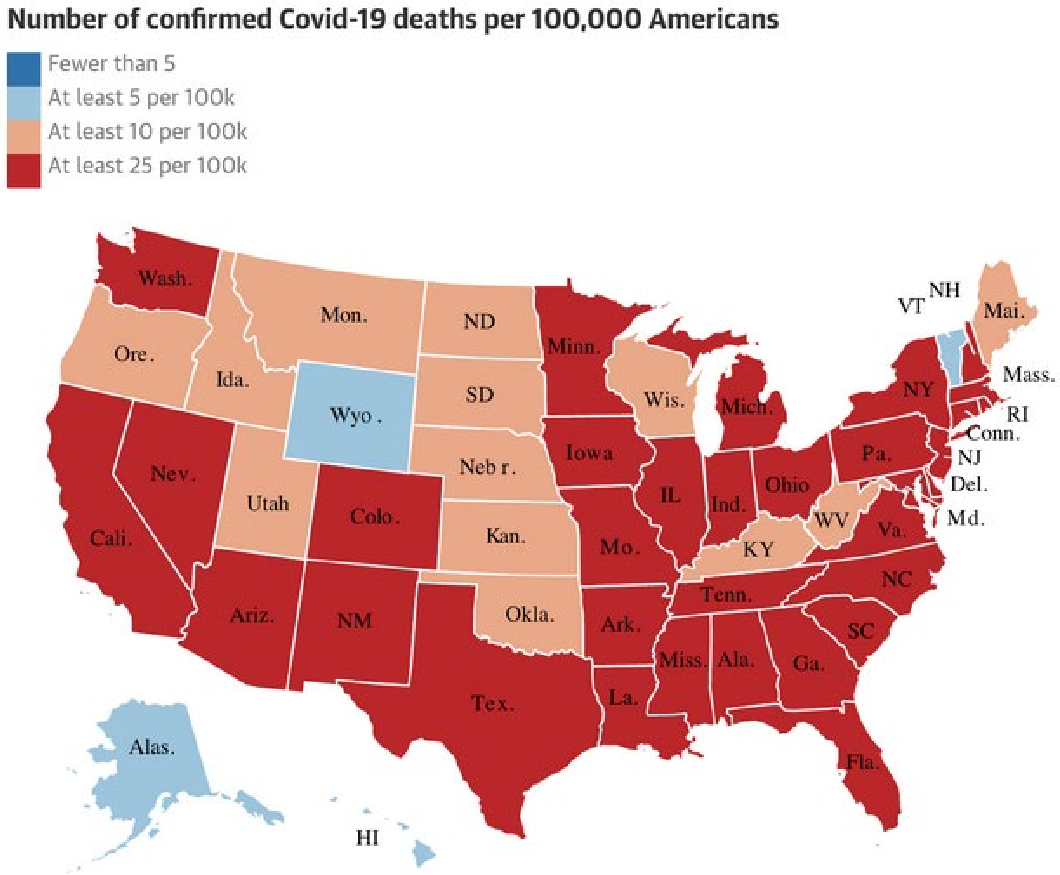
FIGURE 13. Estimated Mean Annual Ratio of Actual Evapotranspiration (ET) to Precipitation (P) for the Conterminous U.S. for the Period 1971-2000. Estimates are based on the regression equation in Table 1 that includes land cover. Calculations of ET/P were made first at the 800-m resolution of the PRISM climate data. The mean values for the counties (shown) were then calculated by averaging the 800-m values within each county. Areas with fractions >1 are agricultural counties that either import surface water or mine deep groundwater.

Another colormap pitfall

I see a large number of graphs that use **diverging colormaps** with diverging points that make no sense, either because:

- The data are not diverging
- The diverging point is random

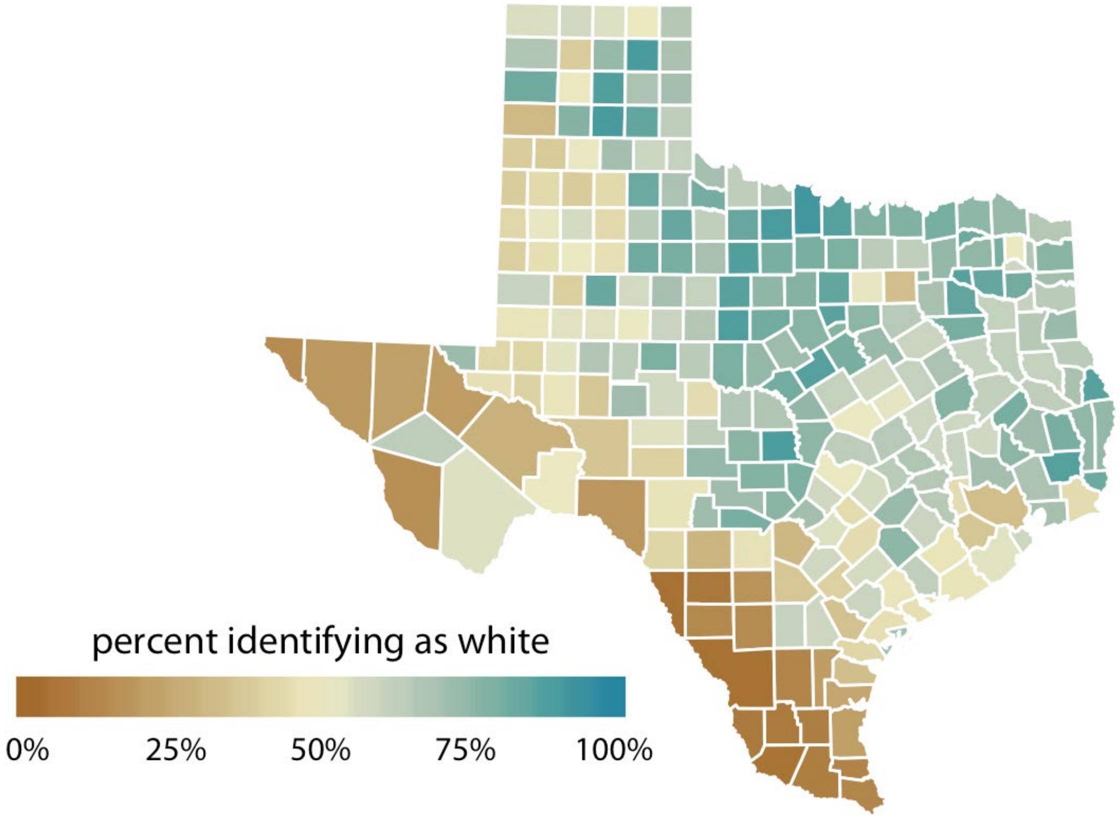
Red-Blue is a popular choice, but the graph on the right does not need to be diverging.



Another colormap pitfall

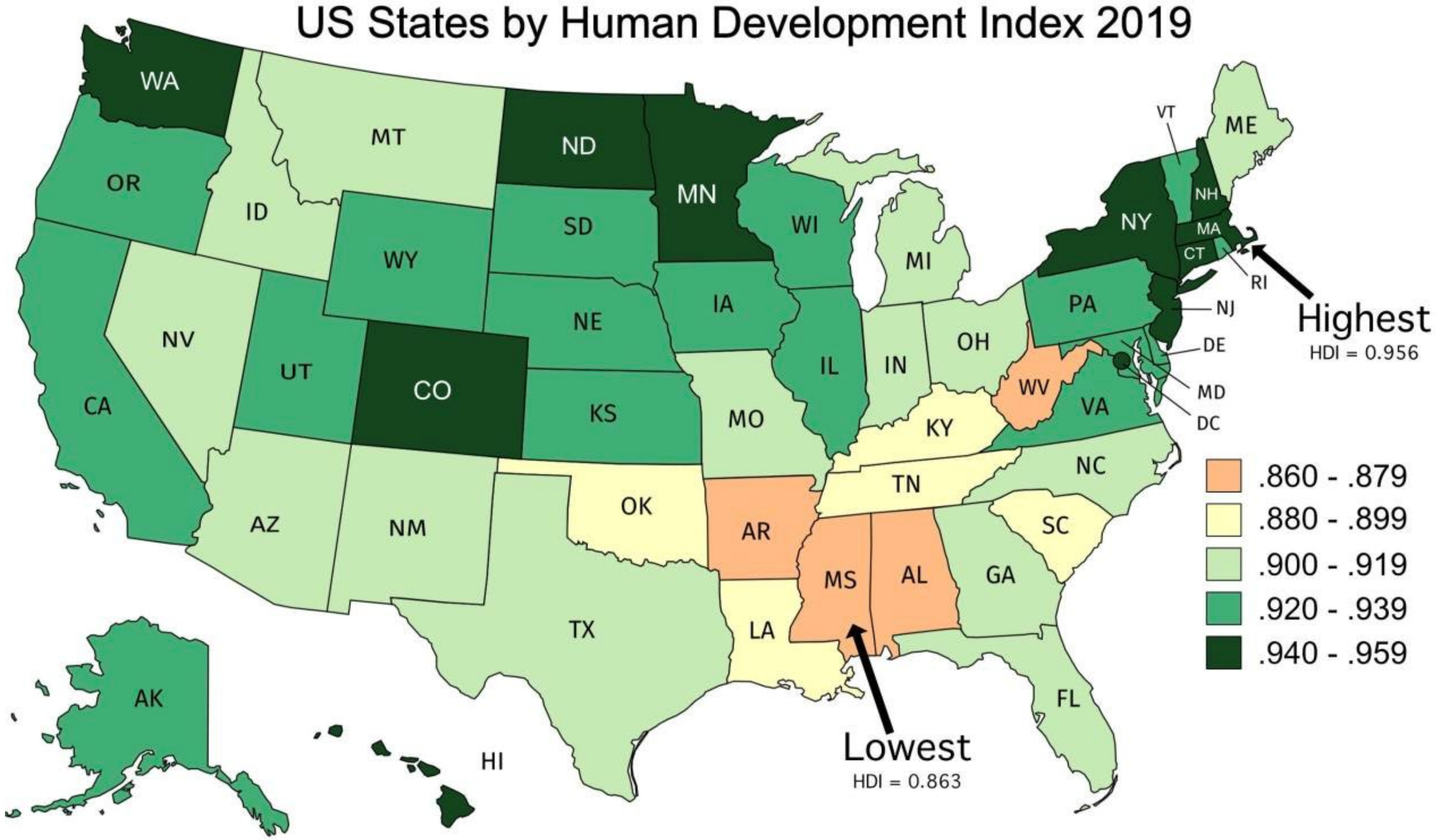
In fact, making it a diverging colormap completely changes the interpretation.

What is the value being highlighted in the graph on the right?



Another colormap pitfall

How about here?

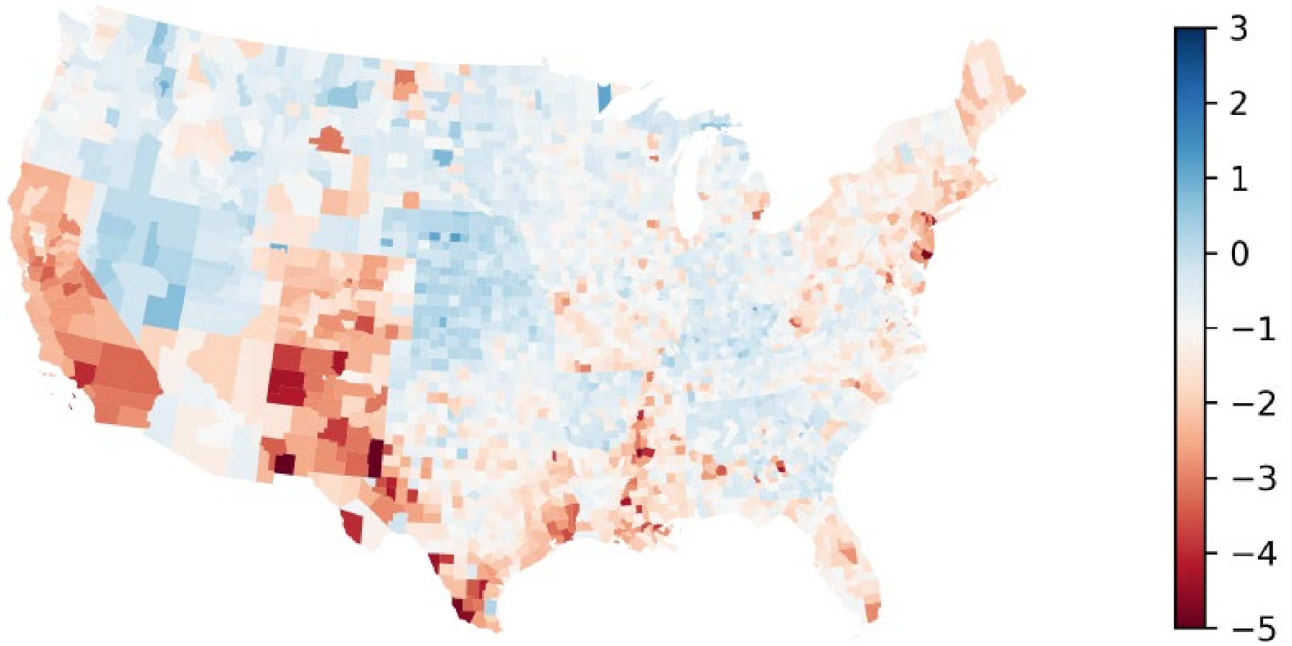


Another colormap pitfall

Another common mistake: an arbitrary diverging point.

What do you conclude about the graph on the right?

Figure 6: *Change in County Unemployment Rates between 2021 and 2022*

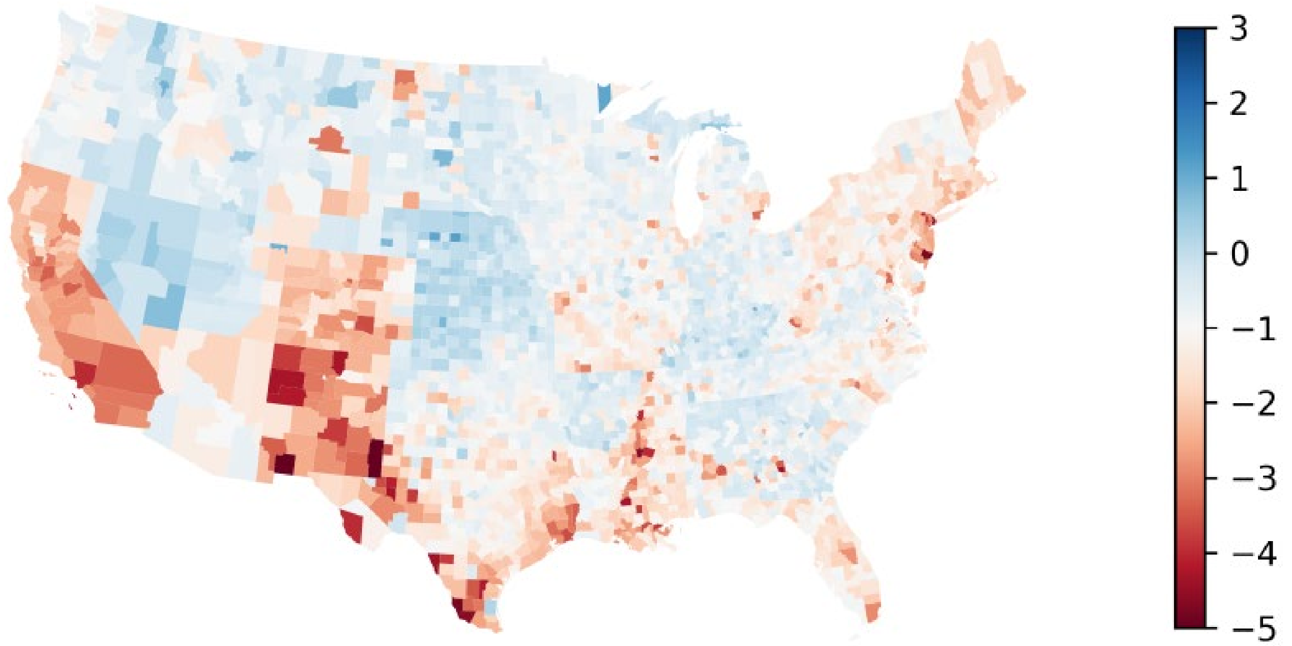


Another colormap pitfall

Using -1 as the diverging point colored made roughly half the counties red and half the counties blue.

What if we made 0 as the diverging point?

Figure 6: Change in County Unemployment Rates between 2021 and 2022



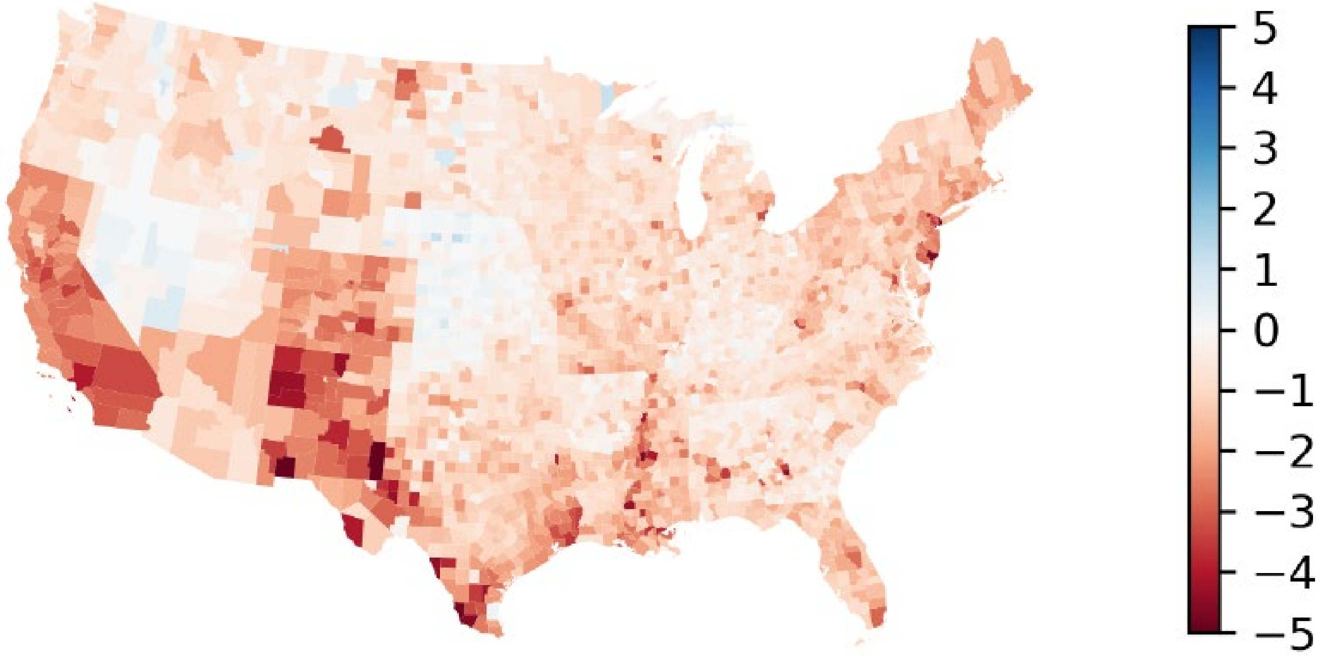
Another colormap pitfall

Using -1 as the diverging point colored made roughly half the counties red and half the counties blue.

What if we made 0 as the diverging point?

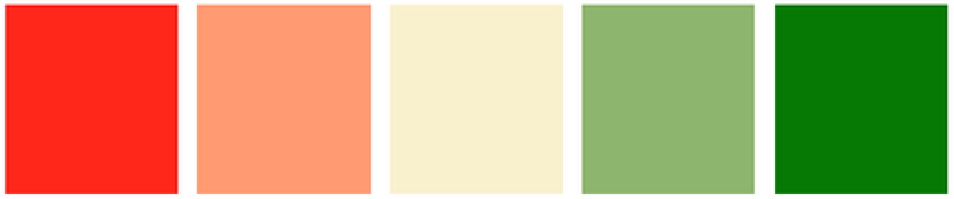
In reality, 98% of counties saw a drop in unemployment.

Figure 6: Change in County Unemployment Rates between 2021 and 2022



One more colormap pitfall

What’s the problem with the colormap to the right?



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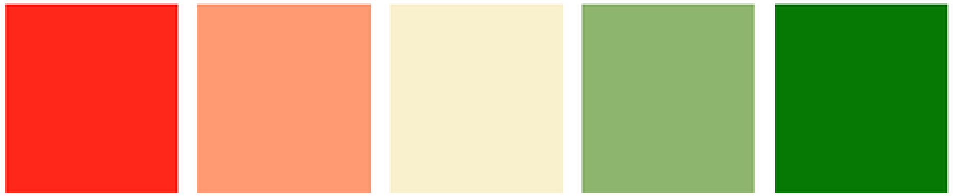

This is what it looks like to people that are R/G colorblind.

One more colormap pitfall

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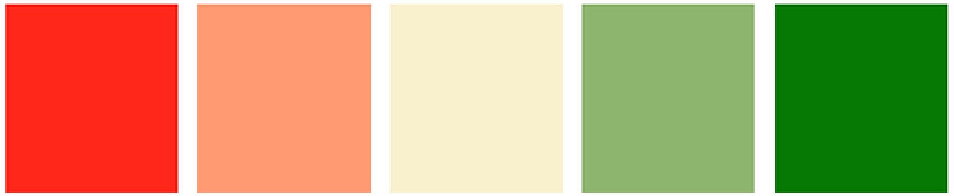
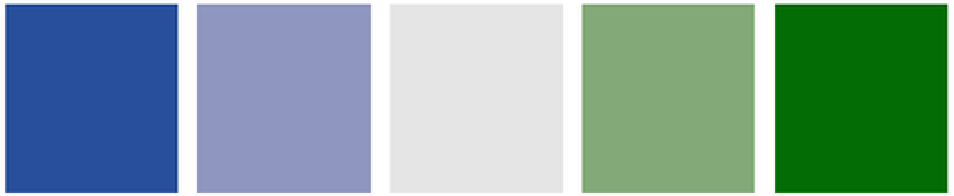
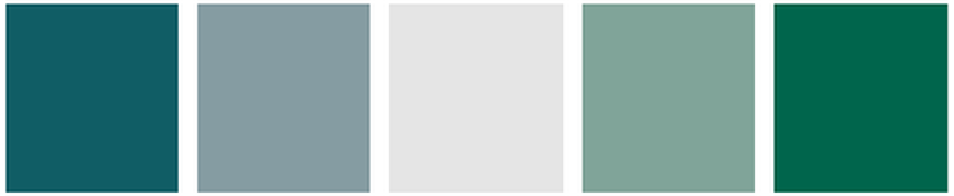


This is what it looks like to people that are R/G colorblind.

Similarly, this is what blue green can look like to some people:





One more colormap pitfall

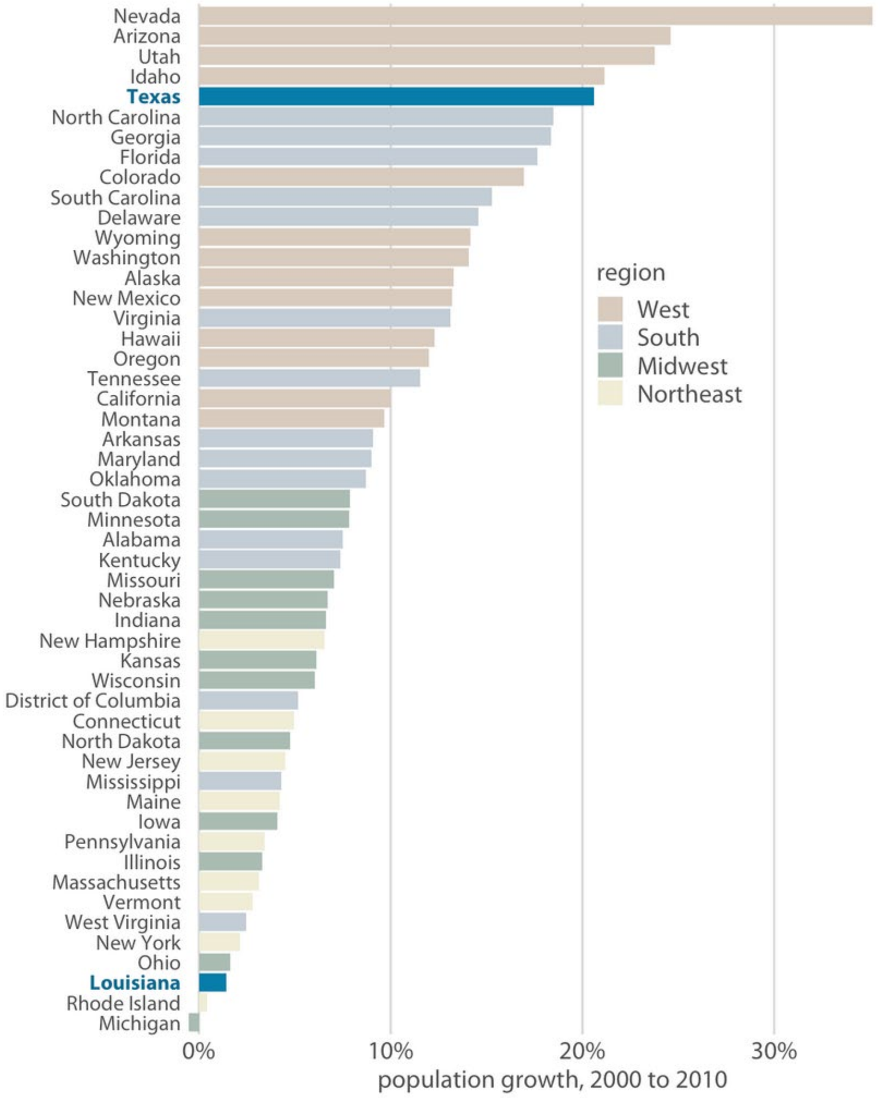
In most graphing packages, the default is something that can be seen with any visual deficiency.

In matplotlib, the default is “viridis” which was developed by python programmers and works even in gray scale

Color to highlight

Using color sparingly is a good way of making things stick out from figures.

Sometimes this means using a colormap but setting the alpha parameter low to make some colors more transparent.



Color in context

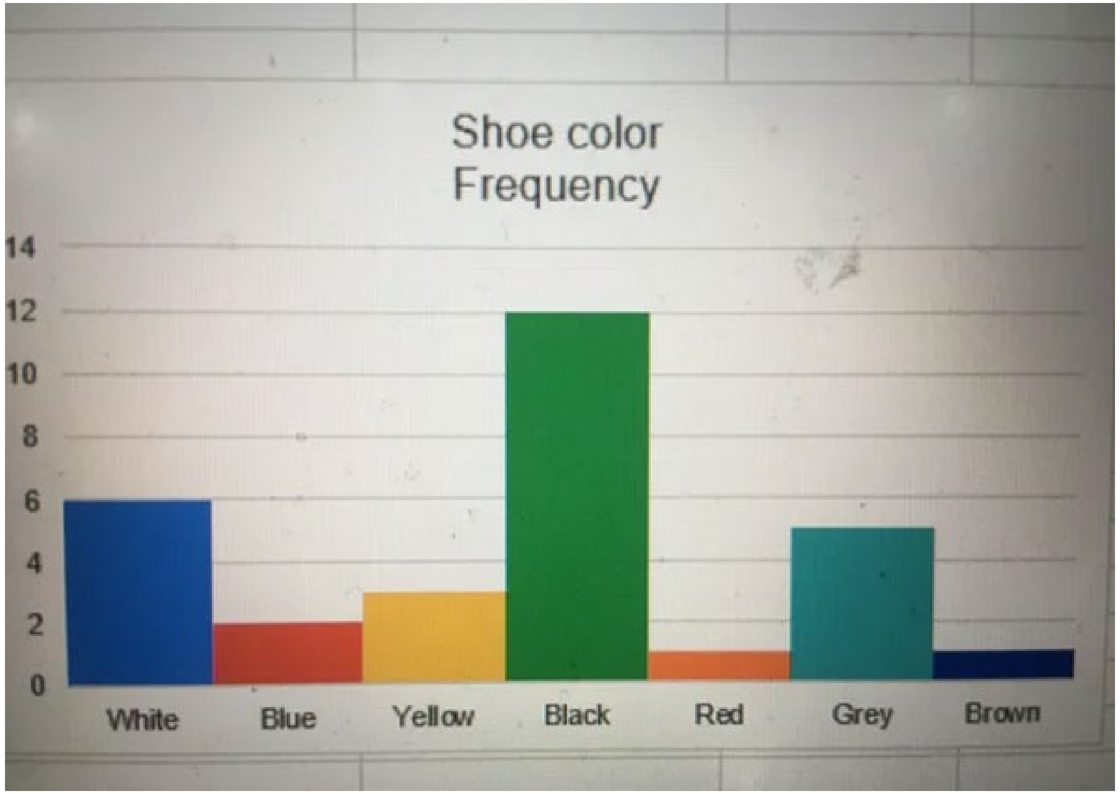
The meaning or intuition of certain colors can have a large impact on your graph.

In US context, some common ones:

- Red: negative, stop, heat
- Green: money, plants, positive
- Blue: cool, water

Colormaps tend to want to go:

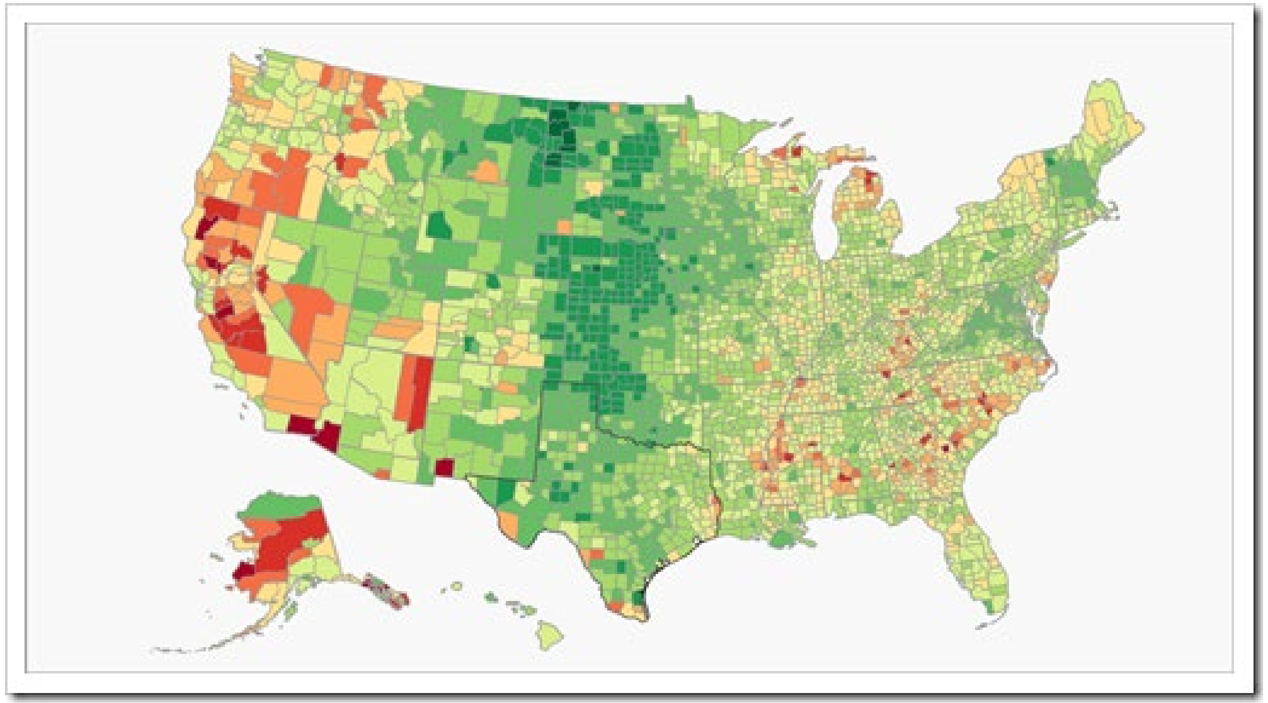
Lighter color = higher number



Color in context

Culture and politics also have an impact.

When you see the figure on the right, what do you interpret the colors as?

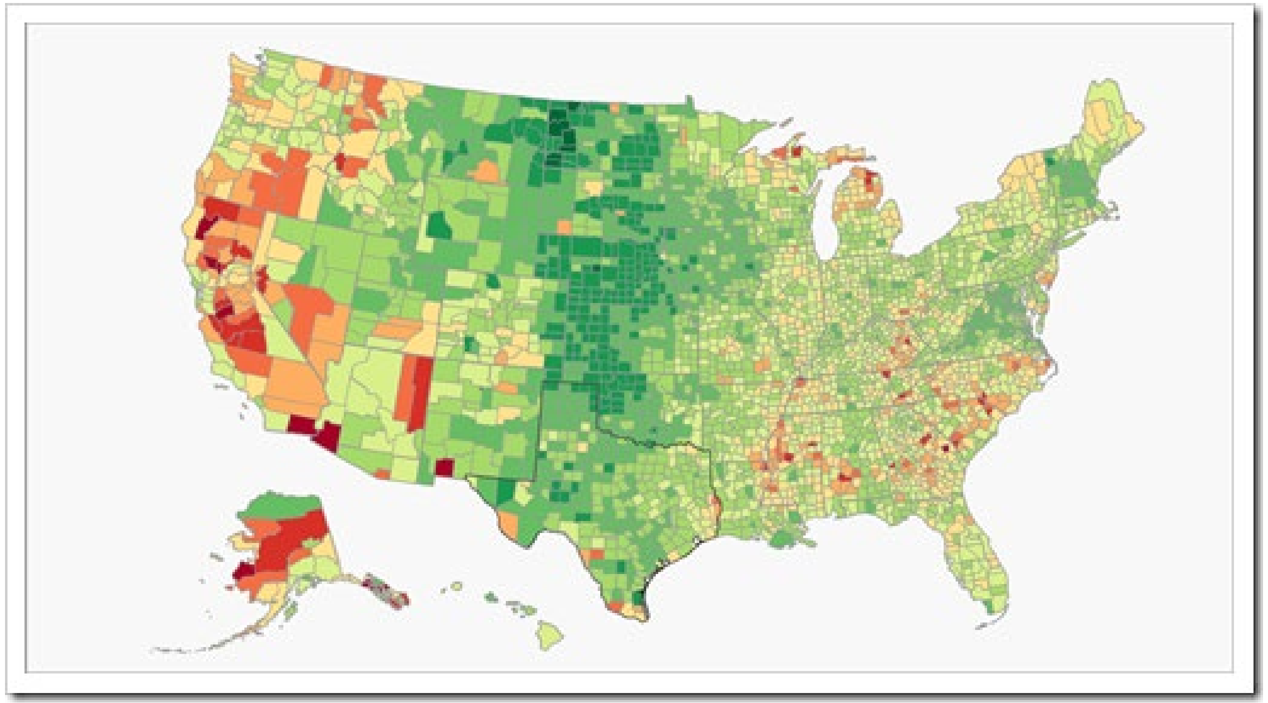


Color in context

Culture and politics also have an impact.

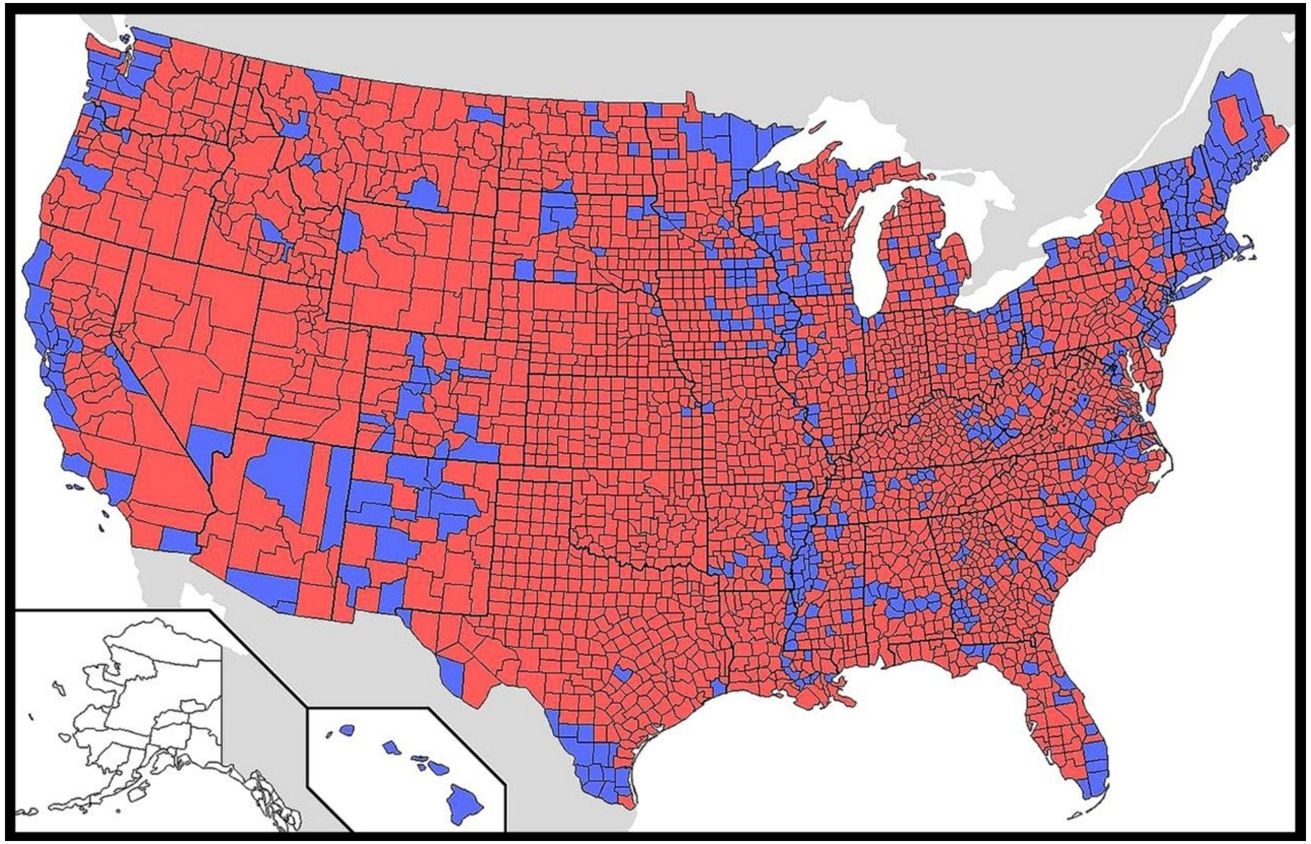
When you see the figure on the right, what do you interpret the colors as?

Answer: red is high unemployment



Color in context

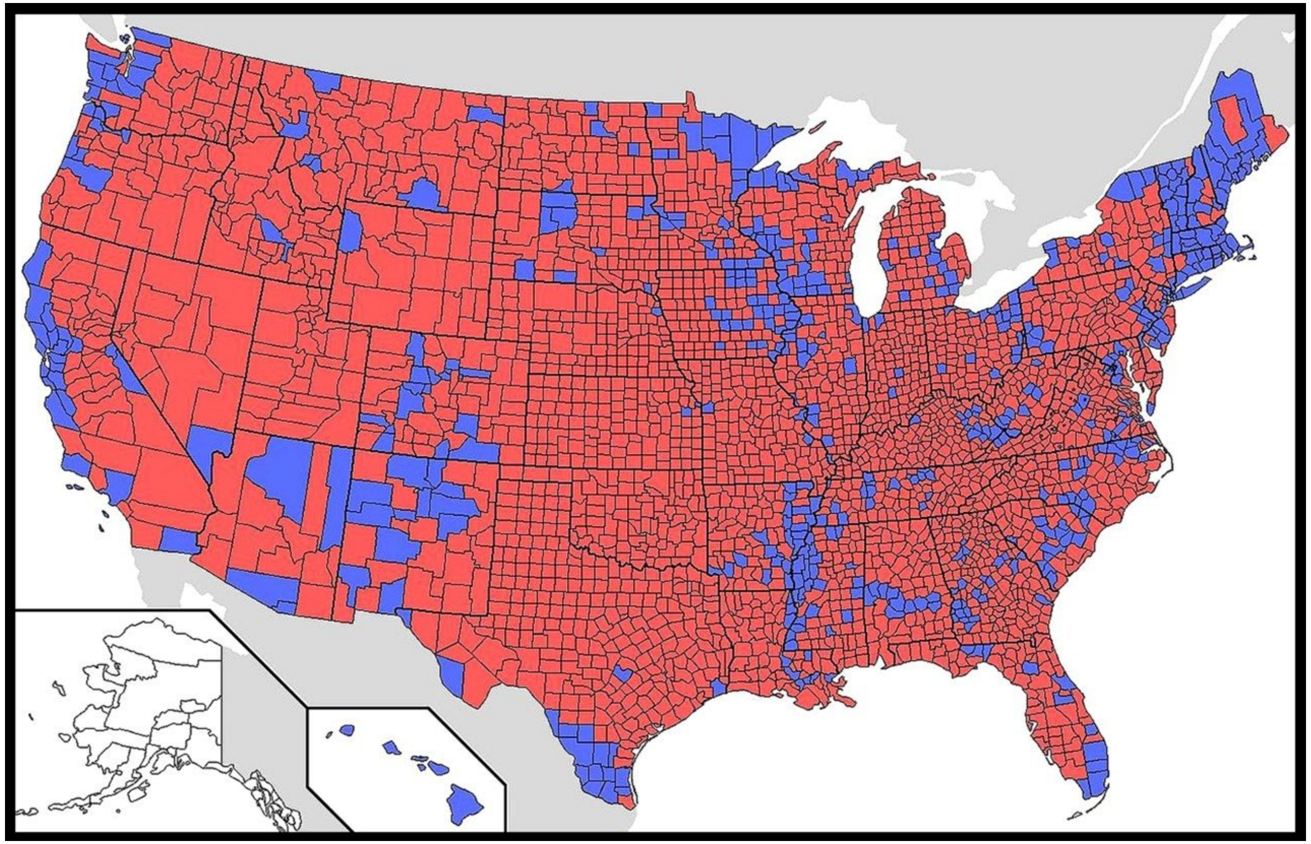
What about this one?



Color in context

What about this one?

In the US, red and blue carry political connotations

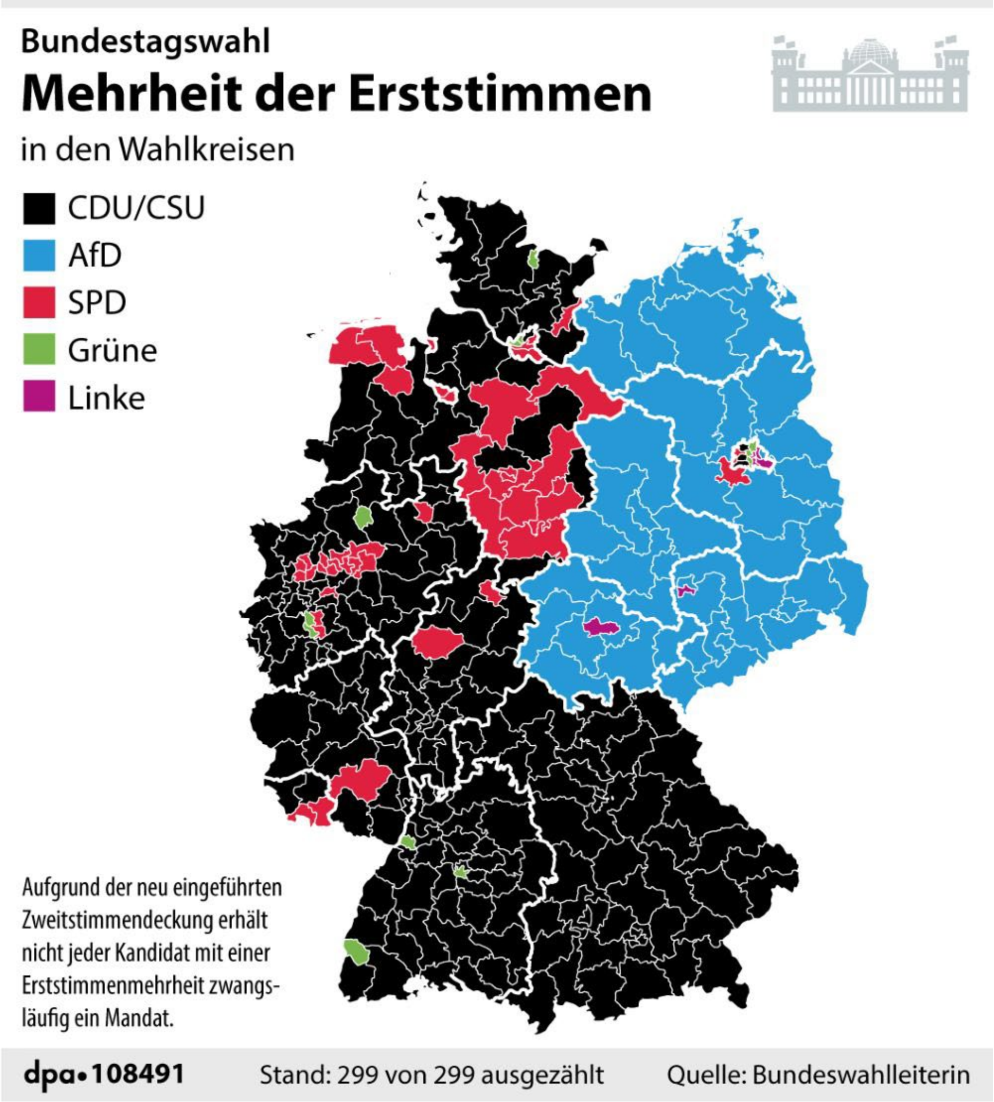


Color in context

What about this one?

In the US, red and blue carry political connotations.

In another country, colors can map to politics very differently



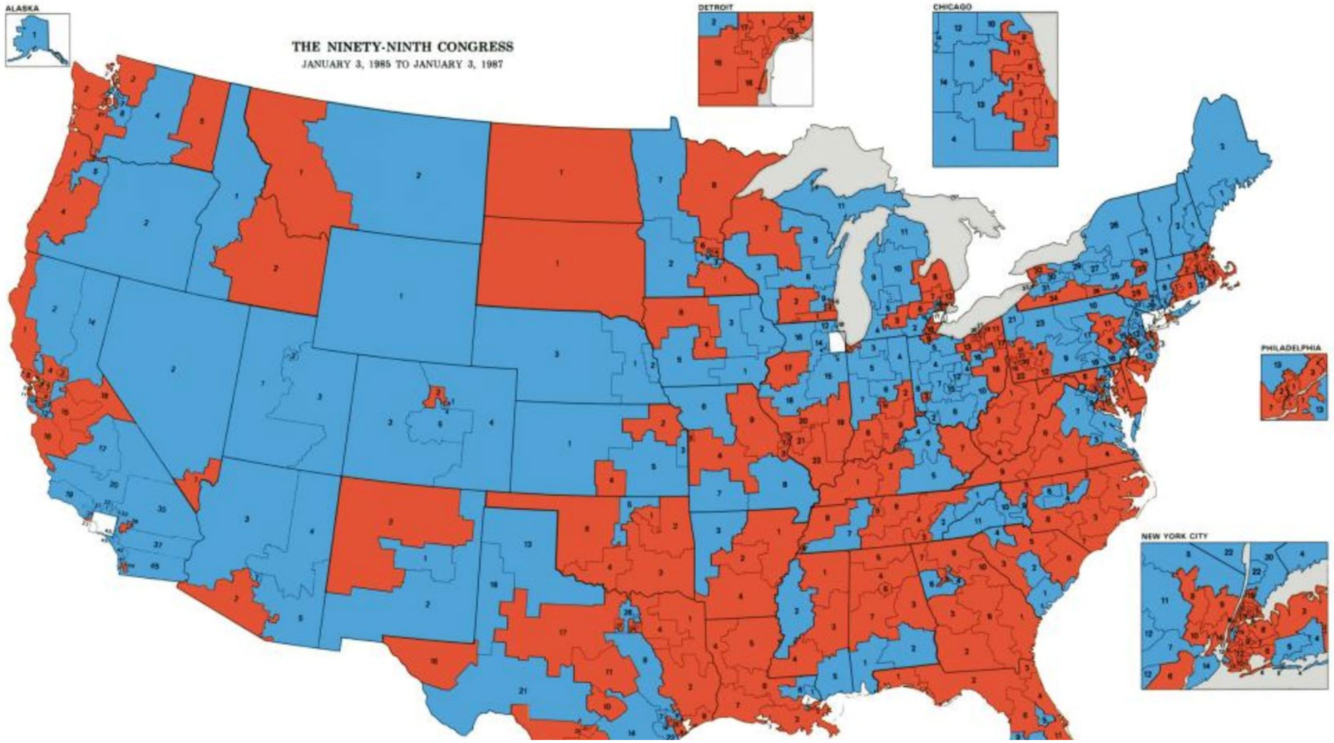
Color in context

What about this one?

In the US, red and blue carry political connotations.

In another country, colors can map to politics very differently

Even US in the past!



Green is positive, red is negative

Dow Jones Industrial Average

As of April 21, 2026 • 2:55 PM CDT

49,158.98 USD +776.59 (1.61%) ↑

1D 5D 1M **YTD** 1Y

49,158.98 USD -73.98 (0.15%) ↓

1D **5D** 1M YTD

Green is positive, red is negative, **but not in Asia!**

Dow Jones Industrial Average

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The “Ick” Test

Due to certain associations, poor use of color can sometimes lead to… icky results.

The “Ick” Test: if your visualization makes someone go “ew,” you have made a mistake.

An example:

The "Ick" Test

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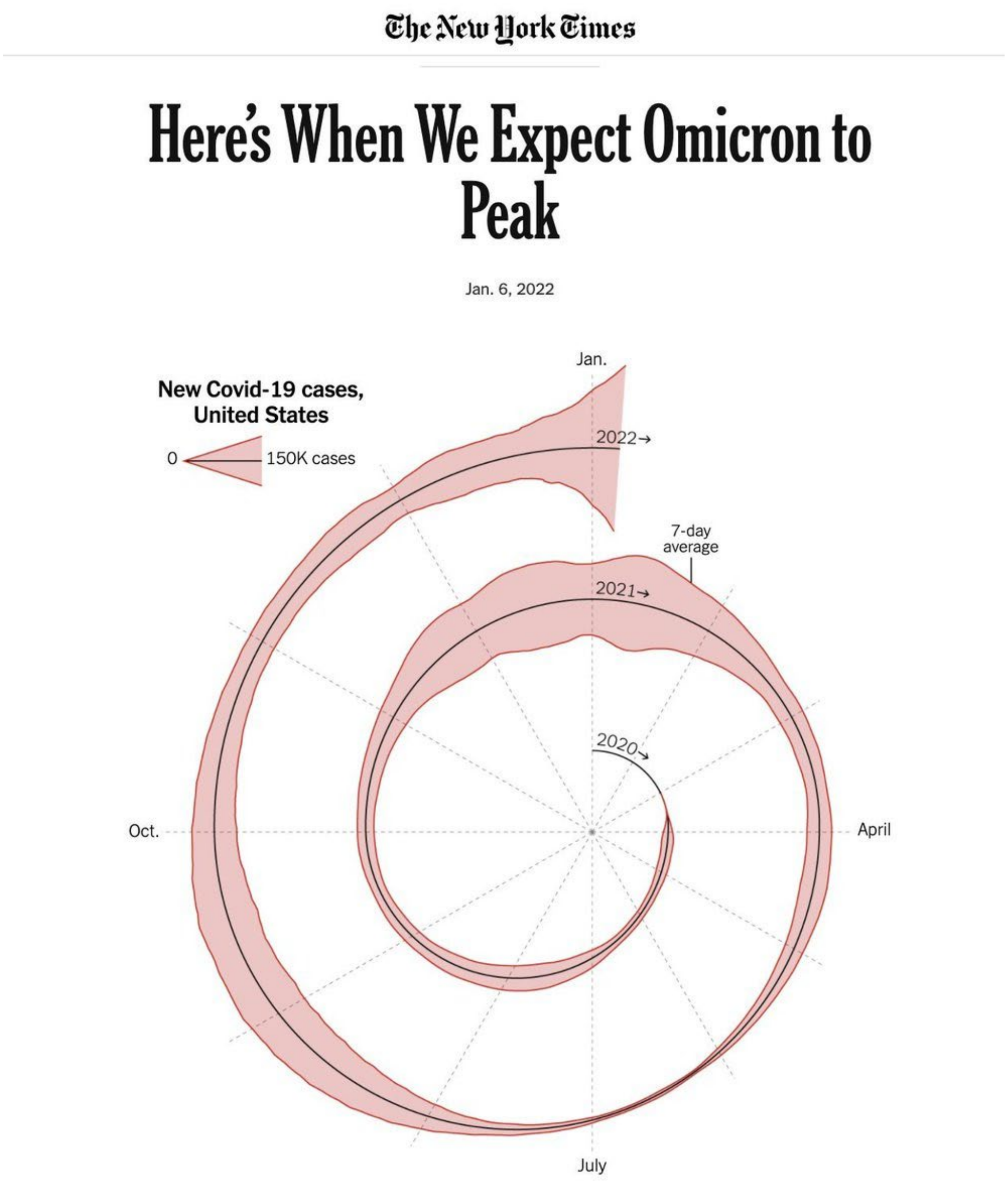
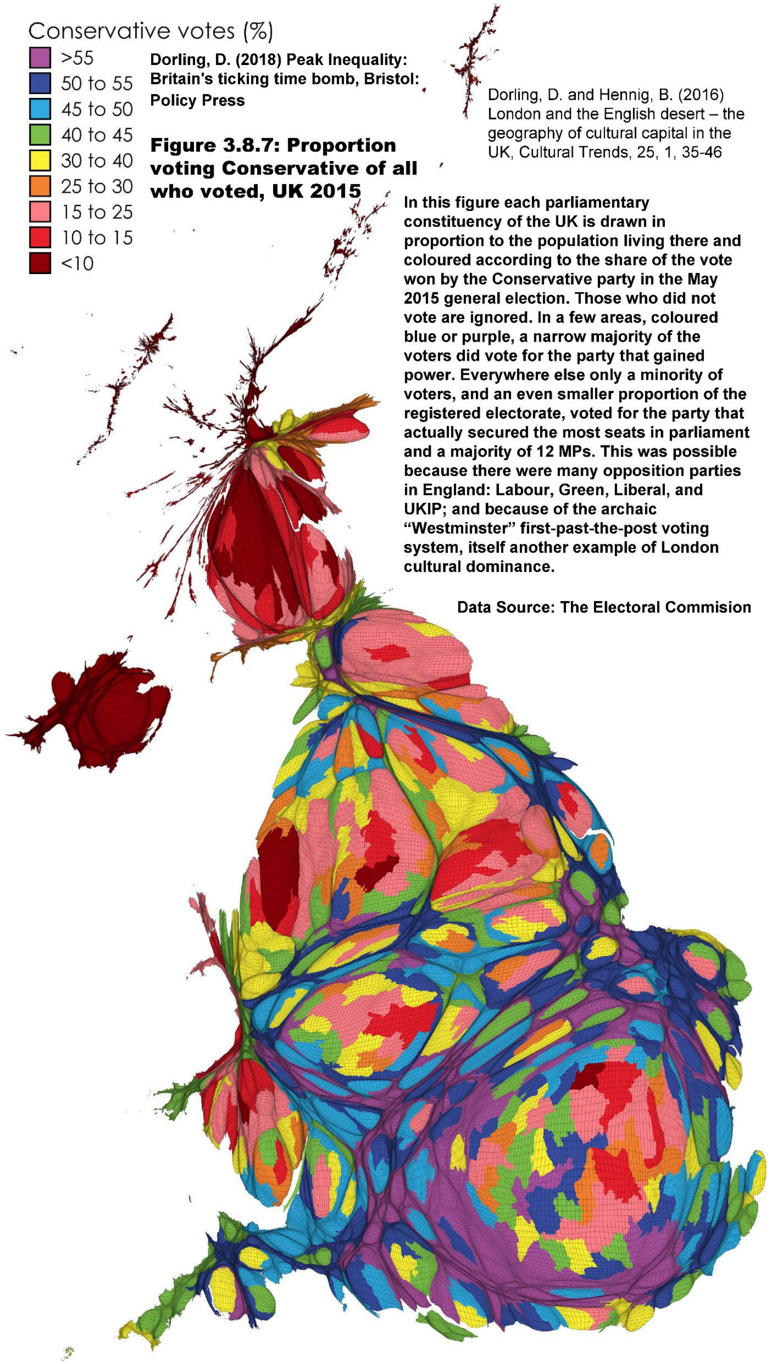
An example:



The “Ick” Test

Perhaps the most egregious example of “ick,” which won “worst data visualization” in my class for several years.

It retains its title even to this day.



Want more? Dig into these two sources!